

The Macro Transcension Hypothesis: Spinning Black Holes in Dense Stellar Clusters as Thermodynamic Attractors for Advanced Civilizations, with Omega Centauri as an Observational Test Bed

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Abstract

Most proposed resolutions of the Fermi paradox assume that long-lived technological civilizations either expand outward, perish, or deliberately hide. We develop a fourth alternative, the *Macro Transcension Hypothesis* (MTH): that civilizations which optimize for long-term computation are driven by thermodynamics, rather than preference, toward a specific class of astrophysical environment, namely rapidly spinning massive black holes embedded in dense, old stellar systems, and that the migration and its endpoint are both electromagnetically quiet. The MTH extends the transcension hypothesis of [Smart \(2012\)](#) from planet-scale “inner space” to macroscopic black-hole infrastructure, and differs from the aestivation hypothesis of [Sandberg et al. \(2016\)](#) in requiring no waiting strategy: the relevant free-energy and entropy-disposal advantages are available now. We quantify the case in four steps: (i) thin-disk accretion onto a Kerr black hole releases 5.7–42 per cent of rest-mass energy ([Bardeen et al., 1972](#); [Novikov and Thorne, 1973](#)), versus 0.7 per cent for hydrogen fusion, while magnetically arrested disks extract additional spin energy at effective efficiencies exceeding 100 per cent of accreted rest mass ([Tchekhovskoy et al., 2011](#)); (ii) by the generalized second law ([Bekenstein, 1974](#)), an event horizon is a thermodynamically ideal entropy sink, and a worked delivery budget (Section 4.2) shows the realized erasure cost lands a factor of $\sim 10^6$ – 10^9 below the CMB-limited Landauer floor once carrier propagation and delivery overheads are charged, with little radiated waste heat; (iii) the Bekenstein–Hawking entropy of a $\sim 2 \times 10^4 M_\odot$ black hole corresponds to $\sim 10^{86}$ bits, exceeding any material archive; and (iv) per unit of harvested mass, this architecture outperforms complete fusion of the same fuel by a factor of ~ 11 –85, outperforms Dyson-type stellar harvesting without star lifting by ~ 80 –600 in lifetime energy yield (~ 11 –85 against a star-lifting economy), and exceeds a solar Dyson swarm by a factor of $\sim 10^9$ (6.6×10^8 at $2 \times 10^4 M_\odot$) in instantaneous Eddington-limited power.

We then identify Omega Centauri (NGC 5139), a $\sim 4 \times 10^6 M_\odot$, ~ 12 -Gyr-old stripped dwarf-galaxy nucleus hosting the nearest strong candidate intermediate-mass black hole ([Häberle et al., 2024b](#)), as the most observationally accessible system satisfying the MTH selection criteria, and present a falsification framework built on six instrument-matched tests: accretion-luminosity limits (JWST, ATCA), mid-infrared waste-heat limits, LISA mass and spin measurements (contingent on a compact-object inspiral occurring in-band during the mission), millisecond-pulsar timing, stellar proper-motion

accelerations, and neutrino burst searches (KM3NeT). Current data, including the unresolved tension between a $\geq 8,200 M_{\odot}$ kinematic lower bound (Häberle et al., 2024b) and a $\lesssim 6,000 M_{\odot}$ pulsar-timing upper bound (Bañares-Hernández et al., 2025) and the complete electromagnetic silence of the central object (Mahida et al., 2026; Chen et al., 2025), are consistent with both the MTH and the more parsimonious gas-starvation null hypothesis; we state explicitly which forthcoming observations would discriminate between them, and which would falsify the MTH outright. The hypothesis is offered in the falsificationist tradition of Sandberg et al. (2016) and Dvali and Osmanov (2023): a speculative but physically grounded working model whose value lies in the concrete observational program it motivates.

Keywords: Fermi paradox; technosignatures; SETI; intermediate-mass black holes; Omega Centauri; NGC 5139; Blandford–Znajek mechanism; Landauer limit; transcension; aestivation

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1 Introduction

The Fermi paradox, in its modern form (Hart, 1975; Tipler, 1980; Webb, 2015; Ćirković, 2018; Forgan, 2019), rests on an expansionist premise: that at least some technological civilizations should grow outward across interstellar distances, and that such growth, sustained for even a small fraction of galactic history, would be conspicuous. The premise is quantitatively robust. Self-reproducing probe architectures permit galaxy-scale colonization in 10^6 – 10^8 years (Tipler, 1980; Freitas and Gilbreath, 1982), intergalactic expansion is energetically cheap for a mature civilization (Armstrong and Sandberg, 2013), and surveys for the waste heat of large-scale energy harvesting have returned null results across $\sim 10^5$ galaxies (Wright et al., 2014; Griffith et al., 2015). The observed silence therefore demands either that technological life is extremely rare, that it reliably destroys itself, or that the expansionist premise is wrong.

A minority tradition in the literature attacks the premise itself. Smart (2012) proposed the *transcension hypothesis*: that the developmental trajectory of intelligence points not outward but inward, toward ever denser, faster, and more efficient configurations of matter and energy (“STEM compression” of space, time, energy, and matter), terminating, in Smart’s telling, at black-hole-scale densities and an effective exit from the observable universe. Sandberg et al. (2016) proposed the *aestivation hypothesis*: that civilizations which value computation should defer it to the far future, when the cosmic microwave background (CMB) has cooled and each erased bit costs less free energy, and should therefore be dormant now. Both proposals share a key insight, that the thermodynamics of computation is the correct lens through which to predict the behaviour of mature intelligence, and both have been criticized on physical grounds. Smart’s mechanism for leaving the universe is not specified in testable form, and the aestivation argument was shown by Bennett et al. (2019) to rest on a thermodynamic error: free energy not harvested now is largely lost, not banked, so a computation-maximizing civilization should collect resources now even if it spends them later.

This paper develops a third member of that family, which we call the *Macro Transcension Hypothesis* (MTH). Its core claim is that the inward trajectory identified by Smart has a concrete, macroscopic, observationally accessible destination in ordinary astrophysics: rapidly spinning massive black holes embedded in dense, dynamically old stellar systems. No new physics, no exit from the universe, and no multi-gigayear dormancy are required. The advantages that drive the migration (accretion power at up to ~ 60 times the mass-energy efficiency of fusion, an event horizon serving as an entropy sink of unlimited capacity, information storage at the Bekenstein–Hawking density, and gravitational time dilation as a controllable computational resource) are all available in the present epoch, and all follow from textbook general relativity and black-hole thermodynamics (Misner et al., 1973; Bardeen et al., 1972; Bekenstein, 1973, 1974). The hypothesis predicts that the most advanced civilizations remain present but electromagnetically invisible, because maximum computational efficiency, viewed from outside, is electromagnetically quiet.

The MTH would remain idle philosophy if it did not select targets. The selection criteria (a massive black hole of modest depth, intermediate rather than supermassive; a dense retinue of low-mass stars as fuel; dynamical age sufficient for any migrating civilization to have arrived and matured; and a quiescent electromagnetic environment) converge on a short list of Galactic systems, at the top of which sits Omega Centauri (ω Cen, NGC 5139): the Milky Way’s most massive globular cluster, widely interpreted as the stripped nucleus of an accreted dwarf galaxy (Hilker and Richtler, 2000; Bekki and Tsujimoto, 2019; Ibata et al., 2019), host to $\sim 10^7$ stars of mean age 12.08 ± 0.01 Gyr (Clontz et al., 2024), and, since the detection of seven fast-moving stars within its central 3 arcsec, host to the strongest current candidate for an intermediate-mass black hole (IMBH) in the Galaxy (Häberle et al., 2024b). That candidate is simultaneously the subject of a sharp observational controversy: combined stellar-kinematic and millisecond-pulsar-timing analysis places a 3σ upper limit of $\sim 6,000 M_\odot$ on any central point mass

(Bañares-Hernández et al., 2025), formally inconsistent with the kinematic lower bound. And it is electromagnetically dark to remarkable depth: neither ~ 170 hours of ATCA radio integration (Mahida et al., 2026), nor JWST infrared photometry (Chen et al., 2025), nor a ~ 290 ks Chandra exposure (Haggard et al., 2013) detects any accretion signature. Every element of this situation (a contested central mass, perfect electromagnetic silence, an imminent decisive instrument in LISA) makes ω Cen a natural laboratory for testing inward-migration hypotheses, entirely independent of whether the MTH is true.

1.1 Claims

Because the argument mixes established physics with speculative inference, we state the epistemic register of each claim explicitly. The paper makes three claims of different strength:

1. **(Established physics plus a comparative synthesis.)** Spinning black holes in dense old clusters are, by a wide quantitative margin, the most thermodynamically favourable environments in the present-day universe for long-term, large-scale computation. The ingredient results are established physics; the optimality claim is a synthesis over a stated comparison set (fusion, Dyson-type stellar harvesting, and cold-space computing), defended quantitatively in Section 4 using standard results (Novikov and Thorne, 1973; Blandford and Znajek, 1977; Bekenstein, 1974; Landauer, 1961; Lloyd, 2000), and is independent of any claim about extraterrestrial intelligence. The ranking of cluster IMBHs above supermassive holes rests partly on strategic rather than thermodynamic criteria (hazard exposure, contested resources), which we treat separately in Section 5.4.
2. **(Speculative hypothesis.)** If long-lived technological civilizations exist and even a subset optimizes for computational capacity, the environments of claim 1 act as attractors, and the resulting migration resolves the Fermi paradox without rare-Earth or doomsday assumptions. This is the MTH proper, stated formally in Section 3. We regard it as less parsimonious than the null hypothesis (technological life is rare or absent) and say so.
3. **(Observational program.)** Whether or not claim 2 is true, it generates instrument-matched, time-bounded, falsifiable predictions at ω Cen, several of which will be adjudicated by existing or funded instruments within ~ 15 years. This program is presented in Section 7.

We do not claim that ω Cen hosts a civilization; we do not claim the IMBH is confirmed (it is not); and we do not claim the observed electromagnetic silence is evidence for the MTH, since it is equally consistent with the simpler gas-starvation explanation that we adopt as the null hypothesis throughout.

1.2 Structure of the paper

Section 2 situates the MTH in the literature. Section 3 states the hypothesis formally as four postulates. Section 4 develops the quantitative thermodynamic and computational case. Section 5 reviews ω Cen and its IMBH evidence, including the current mass tension. Section 6 sketches, at the level of explicitly labelled speculation, the staged architecture by which a migrating civilization would exploit such a system. Section 7 presents the falsification framework and observational roadmap. Section 8 addresses objections, and Section 9 concludes.

2 Relation to prior work

2.1 The expansionist mainstream

The canonical statements of the Fermi paradox (Hart, 1975; Tipler, 1980) treat interstellar expansion as the default behaviour of technological life, an assumption sharpened by Armstrong and Sandberg (2013), who showed that a single mature civilization could seed every galaxy within several billion light-years using a fraction of one star’s output. The “grabby aliens” model of Hanson et al. (2021) quantifies the selection effect: if loud (expanding) civilizations exist at any appreciable rate, they dominate the future light cone, and our own early arrival is informative about their density. Observational programs built on the expansionist premise (radio and optical SETI, Tarter 2001; waste-heat surveys, Dyson 1960; Wright et al. 2014; Griffith et al. 2015; and the broader technosignature framework, Wright et al. 2022; Socas-Navarro et al. 2021; Lingam and Loeb 2021) have produced consistent null results. Within the expansionist frame, these nulls force the unattractive trilemma of rarity, doom, or concealment.

2.2 The inward-migration family

The MTH belongs to a smaller family of proposals holding that mature intelligence migrates down thermodynamic gradients rather than out across space.

Transcension. Smart (2012) argued that the developmental history of complexity on Earth exhibits sustained compression in space, time, energy, and matter (“STEM compression”), and extrapolated to a future in which civilizations engineer environments of black-hole density and disappear from the observable universe. The MTH adopts the compression premise but rejects the terminal step: nothing in known physics permits or requires an exit from the universe, and a hypothesis terminating in unobservable territory cannot be tested. The MTH instead halts the inward trajectory at a macroscopic astrophysical object, an existing massive black hole, where every claimed advantage is calculable and several are observable, which is the sense in which the transcension here is macro.

Postbiological migration. The closest precursor to the present proposal is Ćirković and Bradbury (2006), who argued that postbiological civilizations should migrate down Galactic thermodynamic gradients toward the cold outer Galaxy, where computation is cheaper, and that SETI’s apparent failure reflects a search aimed at the wrong locations. The MTH shares the thermodynamic-migration logic and the diagnosis of SETI’s targeting problem, but identifies a different optimum: the relevant gradient, we argue, is set by the conjunction of energy density, entropy-disposal capacity, and storage density rather than by ambient temperature alone, and its maxima are spinning black holes in dense old clusters rather than the Galactic outskirts.

Aestivation. Sandberg et al. (2016) proposed that civilizations defer computation until the CMB cools, gaining a factor of up to $\sim 10^{30}$ in computations per joule, and so should currently be dormant. Bennett et al. (2019) showed the argument neglects the irreversible loss of free-energy resources not collected promptly: stars burn whether or not anyone harvests them, so maximizers should be active now. The MTH avoids this objection by construction: instead of waiting for a colder universe, it imports a colder-than-CMB entropy sink into the present. Waste entropy crossing an event horizon increases horizon area in accordance with the generalized second law (Bekenstein, 1974), at an effective marginal temperature far below the CMB’s 2.7 K (Section 4.2). An MTH civilization harvests aggressively in the present epoch, consistent with the Bennett–Hanson–Riedel optimality argument, while enjoying much of the efficiency gain that motivated aestivation.

Black holes as civilization infrastructure. The conceit is older than SETI’s engagement with it. Penrose (1969) introduced rotational-energy extraction with an advanced-civilization framing, and Misner

et al. (1973, §33.7) formalized it as the garbage-dump thought experiment: a civilization lowers its waste to just outside the horizon, releases it, and recovers up to 29 per cent of the rest mass as work. Thorne (1994) staged the same power station for a popular audience, and Frolov and Novikov (1998, §12) give the textbook treatment of black holes as energy sources for civilizations. The present series is a completion of that thread. Within SETI proper, several authors have treated black holes as energy sources or computational substrates for advanced intelligence: Inoue and Yokoo (2011) proposed Dyson-type collectors around an accreting supermassive black hole; Hsiao et al. (2021) computed the detectability of a “Dyson sphere around a black hole” harvesting disk luminosity; Opatrný et al. (2017) analysed the inverted configuration in which a black hole serves as the cold reservoir of a heat engine run against the warmer sky; Vidal (2011, 2014) argued that black-hole-coupled (“stellivore”) systems are the natural attractor for cosmic intelligence; and Dvali and Osmanov (2023) analysed black holes as quantum information processors for extraterrestrial civilizations, predicting characteristic high-energy neutrino emission from artificial micro-black-hole computers. Lacki and DiKerby (2025) survey the corresponding high-energy SETI opportunity, and the Dyson Minds workshop (Curtis et al., 2026) has since brought this thread into mainstream observational planning, recommending anomaly-detection searches of archival WISE, JWST, and EHT data for black-hole-hosted intelligence, with a focus on supermassive hosts. The MTH synthesizes this thread and adds the two elements it has lacked: a *selection theory* specifying which black holes are preferred and why (intermediate-mass, high-spin, cluster-embedded; Section 3.1), and a named, ranked observational target with a costed test program.

Kardashev and his inversion. The Kardashev scale (Kardashev, 1964) ranks civilizations by total energy throughput, implicitly rewarding outward expansion. Barrow (1998) proposed the complementary inward scale: mastery over ever-smaller scales of structure. The MTH is the astrophysical completion of Barrow’s inversion — the observation that the universe already provides, free of charge, the densest stable structures that inward mastery could aim at, and that the rational endpoint of a Barrow-type trajectory is therefore relocation rather than fabrication. Bradbury (1999) reached a related conclusion from the engineering side: his Matrioshka-brain analysis shows stellar-powered computation is ultimately limited by radiator area and sink temperature, the constraints an event horizon eliminates.

2.3 New content of the hypothesis

To our knowledge this paper is the first to (i) state the inward-migration resolution of the Fermi paradox as a falsifiable selection theory over real astrophysical objects; (ii) integrate the generalized second law, the Landauer bound, Kerr accretion efficiency, Blandford–Znajek extraction, and Bekenstein–Hawking storage into a single quantitative budget for a civilization-scale computer hosted by an existing black hole; and (iii) attach the resulting predictions to a specific system (ω Cen) with instrument-matched falsification thresholds and timelines. Each ingredient exists in the literature; the synthesis, the selection criteria, and the test program are new.

3 The hypothesis

Epistemic status: Sections 3–4 contain, respectively, a speculative hypothesis stated formally, and established physics offered in its support. The physics in Section 4 stands regardless of the hypothesis.

We state the MTH as four postulates.

P1 (Optimization pressure). Among long-lived technological civilizations, a non-empty subset evolves (by competition, self-modification, or design) toward maximizing long-term computational capacity

per unit of controlled resource. (Motivation: computation is the universal currency into which information-processing agents can convert essentially any terminal goal. This is the instrumental-convergence thesis (Omohundro, 2008; Bostrom, 2012), given formal treatment by Benson-Tilsen and Soares (2016), and the same premise adopted by Smart 2012 and Sandberg et al. 2016. What remains open is the step from convergent instrumental goals in a single agent to a population claim over lineages; P1 asserts only a non-empty subset, the weakest form that step permits.)

P2 (Thermodynamic gradient). For any such maximizer, the present-day universe contains a strict gradient of substrate quality whose global optima are rapidly spinning massive black holes embedded in dense, old, low-luminosity stellar systems. The gradient is set by four independent physical quantities: extractable energy per unit fuel mass, entropy-disposal capacity, information-storage density, and control over effective clock rate. (Defended quantitatively in Section 4.)

P3 (Quiet migration). Migration toward and exploitation of such environments is electromagnetically inconspicuous at interstellar distances. Transit uses gram-to-kilogram-scale beamed-sail payloads rather than worldships (Lubin, 2016); construction uses in-situ resources (Freitas and Gilbreath, 1982); and mature operation, by the very efficiency that motivates it (P2), radiates negligible waste heat (Section 4.2).

P4 (Observable residue). The endpoint is nevertheless not perfectly unobservable. It predicts (a) anomalously high spin on the host black hole; (b) suppressed or absent accretion luminosity relative to ambient gas supply; (c) depletion of the loosely bound stellar population over Gyr timescales, observable today as a low-mass stellar mass-function slope in the central ~ 0.1 pc falling below the envelope of collisional N -body models without engineered feeding (González Prieto et al., 2025); and (d) possibly, burst-mode high-energy neutrino emission if micro-black-hole computation of the Dvali and Osmanov (2023) type is employed. These residues define the falsification program of Section 7.

The hypothesis (MTH). *If P1–P3 hold, then the oldest computation-maximizing civilizations in the Galaxy have already relocated to environments of the P2 class; their absence from planetary systems, the radio spectrum, and waste-heat surveys is a selection effect of the thermodynamic gradient; and the appropriate search strategy is targeted scrutiny of the small set of P2-optimal Galactic systems for the P4 residues.*

Three corollaries sharpen the contrast with neighbouring hypotheses. First, against aestivation: the MTH predicts present activity in P2 environments, not dormancy, so it is not vulnerable to the Bennett et al. (2019) objection. Second, against the dark-forest family of concealment hypotheses: silence here is a by-product of efficiency, not a strategic choice, so the MTH requires no game-theoretic coordination among civilizations. Third, against Smart’s original transcension: every MTH process occurs in ordinary spacetime around an ordinary Kerr black hole, so the hypothesis remains permanently coupled to observation.

3.1 Selection criteria over host systems

P2 implies a ranking of real systems. The optimum is set by competing requirements:

1. **Black-hole mass in the intermediate range** (10^3 – $10^5 M_\odot$). Energy extraction (Section 4.1) scales with available fuel and with Eddington-limited throughput ($\propto M$), favouring larger M ; but tidal disruption of solid infrastructure at the innermost stable circular orbit (ISCO) becomes

less constraining at larger M , while two costs grow with M : the Bekenstein–Hawking storage already saturates any plausible need at $10^4 M_\odot$ (Section 4.3), and supermassive holes sit in deep galactic-centre potentials crowded with disruptive astrophysics (Section 5.4). The intermediate range secures every advantage at minimal exposure.

2. **High spin, or spin-up feasibility.** The Blandford–Znajek channel and the deep-ISCO efficiencies require $a_\star \gtrsim 0.9$; a hole of modest spin can be spun up by prograde disk accretion at a calculable mass cost (Thorne, 1974), provided fuel exists (criterion 3).
3. **A dense, old, low-mass stellar reservoir.** Sustained Eddington-limited feeding of a $\sim 2 \times 10^4 M_\odot$ hole consumes of order one solar mass per two millennia (Section 4.1); at ω Cen’s central density the inner parsec supplies $\sim 3 \times 10^4$ stars and the central ~ 10 pc of order 10^6 , at relative velocities of tens of km s^{-1} . The contested status of this fuel is era-relative: it is uncontested by fusion-era economics, but under star lifting the same M dwarfs are prize fuel (fully convective, hence hydrogen-complete, with $\sim 10^{12}$ -yr lifetimes; Scoggins and Kipping 2023), so among computation-maximizing lineages the reservoir is maximally contested, an early-arrival premium developed in the companion economics paper.
4. **Dynamical age and quiescence.** A system $\gtrsim 10$ Gyr old has been reachable by any Galactic civilization arising in the first stellar generations; a relaxed, gas-poor environment minimizes both natural accretion noise and hazards to infrastructure.

Milky Way globular clusters with massive-IMBH candidates satisfy all four criteria simultaneously; ω Cen satisfies them maximally (Section 5). The Galactic Centre fails criterion 1 and 4; isolated stellar-mass black holes fail criterion 3; young massive clusters fail criterion 4. Retention is not the bottleneck: survival analyses of IMBHs in dense clusters find that holes above $\sim 10^3 M_\odot$ are generally retained against dynamical ejection over a Hubble time (Martinez et al., 2026), so if formation occurs at all, the P2 class is populated today.

Because a selection theory that never exhibits its selection invites the charge of unfalsifiability by retreat, Table 1 scores the leading Galactic candidates against the four criteria. The ranking is coarse (the IMBH-evidence column in particular is contested for every entry), but it is the operational content of P2: the class-level kill statement of Section 8 quantifies over exactly these systems.

4 The thermodynamic and computational case

This section defends postulate P2 quantitatively. Throughout, M is the black-hole mass, $a_\star \equiv Jc/GM^2 \in [0, 1)$ the dimensionless spin, $r_g \equiv GM/c^2$ the gravitational radius, and we evaluate fiducial numbers at $M = 2 \times 10^4 M_\odot$, the geometric midpoint of the contested ω Cen range (Section 5). All results in this section are standard physics; no claim about extraterrestrial intelligence is made until Section 6.

4.1 Energy: accretion and spin extraction versus fusion

Thin-disk efficiency. Matter spiralling through a geometrically thin, radiatively efficient disk is released at the ISCO binding energy. For a circular equatorial orbit at radius r around a Kerr hole, the specific energy at the ISCO gives a radiative efficiency (Bardeen et al., 1972; Novikov and Thorne, 1973)

$$\eta(a_\star) = 1 - E_{\text{ISCO}} = 1 - \sqrt{1 - \frac{2}{3} \frac{r_g}{r_{\text{ISCO}}(a_\star)}}, \quad (1)$$

Table 1: The P2 selection criteria applied to the leading Galactic candidate systems. Symbol columns (current IMBH evidence; inferred or claimed mass versus the preferred 10^3 – $10^5 M_\odot$ range; low-mass fuel reservoir; dynamical age and quiescence): ++ strong, + favourable, ○ neutral or contested, – unfavourable. Quantitative columns: distance and central mass density from [Harris \(1996\)](#); [Baumgardt and Vasiliev \(2021\)](#), rounded to leading digit; $r_{\text{infl}} = GM/\sigma_c^2$ evaluated at a reference $10^4 M_\odot$ and the catalogued central dispersion; best current central-mass constraint with source.

System	Ev.	Mass	Fuel	Age	d (kpc)	ρ_c ($M_\odot \text{pc}^{-3}$)	r_{infl} (pc)	Best current mass constraint ($M_\odot$)
ω Cen	++	++	++	++	5.5	$\sim 3 \times 10^3$	0.10	$\geq 8.2 \times 10^3$ (kinematic) vs. $< 6 \times 10^3$ (timing); contested (Section 5.2) (Häberle et al., 2024b ; Bañares-Hernández et al., 2025)
47 Tuc	○	+	++	++	4.5	$\sim 2 \times 10^5$	0.3	$\sim 2.3 \times 10^3$ timing claim (Kızıltan et al., 2017); not required by later modelling (Mann et al., 2019)
M54	+	+	+	+	26.3	$\sim 10^5$	0.2	$\sim 10^4$ kinematic hint (Greene et al., 2020); Sgr dwarf nucleus
NGC 6388	○	+	+	+	9.9	$\sim 5 \times 10^5$	0.12	kinematic claims disputed; deep radio null, $\lesssim \text{few} \times 10^3$ (Tremou et al., 2018)
NGC 6441	○	+	+	+	11.6	$\sim 4 \times 10^5$	0.13	as NGC 6388; radio null (Tremou et al., 2018)
Terzan 5	–	○	++	○	6.9	$\sim 3 \times 10^5$	0.2	no direct evidence (heavy bulge extinction); radio null (Tremou et al., 2018)
M15	○	○	++	++	10.4	$\sim 10^{6-7}$ (cusp)	0.25	$\lesssim 10^{3-4}$, model-dependent; kinematics equally fit by remnants (Greene et al., 2020)

which rises from $\eta = 1 - \sqrt{8/9} \simeq 0.057$ at $a_\star = 0$ ($r_{\text{ISCO}} = 6 r_g$) to 0.321 (pure geodesic value; the photon-capture-corrected equilibrium efficiency is $\simeq 0.30$) at the [Thorne \(1974\)](#) radiative spin-equilibrium limit $a_\star = 0.998$, and formally to $1 - 1/\sqrt{3} \simeq 0.423$ as $a_\star \rightarrow 1$ (Fig. 1). Hydrogen fusion releases 0.71 per cent of the rest mass burned, but a real star offers only its hydrogen fraction $X_{\text{H}} \simeq 0.7$ to that channel, so complete fusion of a star yields $\simeq 5 \times 10^{-3}$ of stellar mass; an unassisted main-sequence star burns roughly a tenth of its mass over its lifetime, so a Dyson-type collector ([Dyson, 1960](#)) captures a lifetime yield of order 7×10^{-4} . Per unit of fuel mass under the civilization’s control, disk accretion onto a high-spin hole thus outperforms complete fusion of the same mass by a factor of ~ 11 (Schwarzschild) to ~ 85 (extremal), and outperforms Dyson harvesting without star lifting by a factor of ~ 80 – 600 . The comparison set matters, because star lifting changes it: a civilization that strips the envelope and feeds the core accesses close to the full hydrogen inventory ([Criswell, 1985](#); [Matloff, 2017](#); [Scoggins and Kipping, 2023](#)), raising the yield toward the 5×10^{-3} ceiling and collapsing the Dyson disadvantage to the same ~ 11 – 85 as complete fusion. The entropy-sink and storage advantages (Sections 4.2–4.3) are independent of which factor applies.

Power throughput. The sustainable luminosity is bounded by the Eddington limit,

$$L_{\text{Edd}} = \frac{4\pi G M m_p c}{\sigma_T} \simeq 1.26 \times 10^{31} \left(\frac{M}{M_\odot} \right) \text{ W}, \quad (2)$$

giving 1.0×10^{35} , 2.5×10^{35} , and 6.3×10^{35} W at $M = 8.2 \times 10^3$, 2×10^4 , and $5 \times 10^4 M_\odot$ respectively ([Frank et al., 2002](#); [Shapiro and Teukolsky, 1983](#)), some 3×10^8 – 2×10^9 times the bolometric output of the Sun across that mass range (6.6×10^8 at $2 \times 10^4 M_\odot$), and hence of any solar Dyson swarm (Fig. 2). The corresponding Eddington accretion rate, $\dot{M}_{\text{Edd}} = L_{\text{Edd}}/\eta c^2$, is remarkably modest: at $\eta = 0.1$ and $M = 2 \times 10^4 M_\odot$, one solar mass sustains the limit for $\sim 2,300$ years. At ω Cen’s adopted central density ($\sim 3 \times 10^3 M_\odot \text{pc}^{-3}$; Table 3) the central parsec holds $\sim 1.3 \times 10^4 M_\odot$, roughly 3×10^4 stars, or

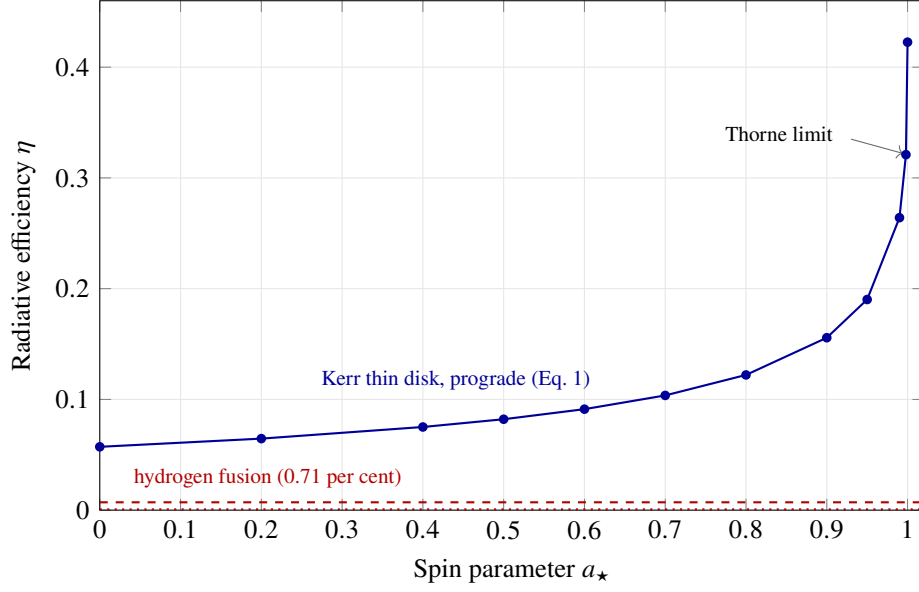


Figure 1: Mass-to-energy conversion efficiency of thin-disk accretion as a function of black-hole spin (Bardeen et al., 1972; Novikov and Thorne, 1973), compared with hydrogen fusion and with the lifetime yield of Dyson-type stellar harvesting. The Thorne (1974) limit $a_* = 0.998$ marks the equilibrium spin reachable by disk accretion alone; the plotted $\eta \approx 0.32$ there is the geodesic value, reduced to ≈ 0.30 once photon capture is included (Thorne, 1974). Curve drawn through exact values of Eq. (1) at the plotted points.

$\sim 3 \times 10^7$ yr of Eddington fuel on its own; the working reservoir is the central ~ 10 pc, whose $\sim 10^6$ stars fuel Eddington-limited operation for 10^{8-9} years using bodies (brown dwarfs, low-mass M dwarfs, white dwarfs) that no fusion-based economy values, at the delivery cost priced in Section 6.

Spin energy and the Blandford–Znajek channel. A Kerr hole stores extractable rotational energy

$$E_{\text{rot}} = \left[1 - \sqrt{\frac{1}{2}(1 + \sqrt{1 - a_*^2})} \right] M c^2, \quad (3)$$

reaching $0.29 M c^2 \approx 1.0 \times 10^{51}$ J for $M = 2 \times 10^4 M_\odot$ at $a_* \rightarrow 1$: a reservoir equal to the entire lifetime fusion output of $\sim 10^7$ Suns, storable indefinitely and tappable on demand. The bracketed quantity is the Christodoulou–Ruffini reducible energy, the difference between M and the irreducible mass M_{irr} (Christodoulou, 1970; Christodoulou and Ruffini, 1971); extraction is reversible only in the limit of vanishing horizon-area increase, and inefficient extraction raises M_{irr} permanently, destroying future capacity, a distinction that matters for the spin-fossil reasoning of Section 7. That this energy is extractable in principle was shown by the ergosphere mechanism of Penrose (1969) and Penrose and Floyd (1971); the astrophysically efficient extraction mechanism is electromagnetic: a horizon threaded by poloidal magnetic flux Φ supplied by an accretion disk acts as a unipolar inductor, driving Poynting-flux jets with power $P_{\text{BZ}} \propto \Phi^2 \Omega_H^2$, where Ω_H is the horizon angular frequency (Blandford and Znajek, 1977). In the membrane paradigm of Thorne et al. (1986) the horizon carries a surface resistivity of 377Ω and the whole configuration reads as a battery with internal resistance, the engineering-facing form of the mechanism and the one relevant to Section 6’s power-station framing. In the magnetically arrested disk (MAD) regime, general-relativistic magnetohydrodynamic simulations find jet efficiencies $\eta_{\text{jet}} \equiv P_{\text{jet}}/\dot{M} c^2 \approx 1.4$ at $a_* \approx 0.99$ (Tchekhovskoy et al., 2011; Tchekhovskoy, 2015): the jet carries more energy than the

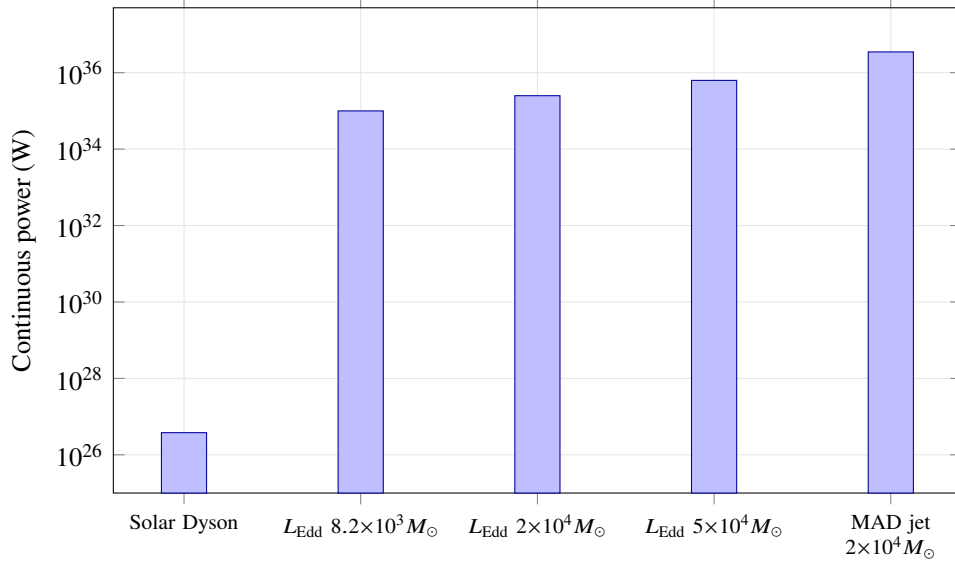


Figure 2: Continuous power available to a civilization from a complete solar Dyson swarm versus Eddington-limited accretion (Eq. 2) onto IMBHs spanning the contested ω Cen mass range, and a magnetically arrested (MAD) Blandford–Znajek jet at $a_{\star} \simeq 0.99$ with $\eta_{\text{jet}} \simeq 1.4$ (Tchekhovskoy et al., 2011), fed at the $\eta = 0.1$ Eddington rate; the two efficiency conventions combine as $P_{\text{jet}} = \eta_{\text{jet}} \dot{M} c^2 = \eta_{\text{jet}} (L_{\text{Edd}}/\eta)$. The IMBH advantage in instantaneous power is roughly nine orders of magnitude.

accreted rest mass, the excess drawn from spin. The BZ scaling has recently been confirmed from first principles by ab-initio particle-in-cell simulations of Kerr magnetospheres (Meringolo et al., 2025), which additionally identify equatorial magnetic reconnection as a supplementary extraction channel (Comisso and Asenjo, 2021; Camilloni and Rezzolla, 2025); one 2026 preprint reports transient MAD states with jet power exceeding accretion power by over two orders of magnitude (Nathanail, 2026, not yet peer-reviewed). For the MTH it suffices that $\eta_{\text{jet}} \gtrsim 1$ is robust across methods: an engineered MAD configuration around a spun-up IMBH is, per unit fuel, the most efficient large-scale power plant permitted by demonstrated physics.

Spin-up economics. A hole of low natal spin is itself improvable. Prograde thin-disk accretion drives $a_{\star} \rightarrow 0.998$ after the hole grows by a factor ≈ 2.2 (2.2024) (Bardeen, 1970; Thorne, 1974); for an initial $2 \times 10^4 M_{\odot}$ hole this costs $\sim 2.4 \times 10^4 M_{\odot}$ of accreted matter (of order 10^{4-5} cluster stars) and, at Eddington throughput, $\sim 3.5 \times 10^7$ years. The investment multiplies all subsequent per-unit-fuel returns by up to ~ 6 (Fig. 1) and unlocks the BZ reservoir. The corollary for observers: spin is a key parameter of the system’s history (Section 7), since the MTH requires engineered systems to approach near-extremal spin; the converse inference is weaker, because some natural formation channels, in particular coherent disk growth in a gas-rich nucleus, also reach high spin (Section 7; Reynolds, 2021).

4.2 Entropy disposal: the horizon as the ultimate cold reservoir

Landauer’s principle sets the minimum thermodynamic cost of irreversible computation: erasing one bit dissipates at least $k_B T \ln 2$ into a reservoir at temperature T (Landauer, 1961; Bennett, 1982; Parrondo et al., 2015), a bound verified experimentally (Bérut et al., 2012). Reversible logic evades the bound for intermediate steps (Bennett, 1973; Toffoli, 1980; Fredkin and Toffoli, 1982; Frank, 2002; Athas et al.,

1994), but error correction, measurement, and memory reuse impose a residual erasure budget on any real computer (Wolpert, 2019). For a conventional deep-space computer the reservoir floor is the CMB at $T_\gamma = 2.7$ K, so the marginal cost is $k_B T_\gamma \ln 2 = 2.6 \times 10^{-23}$ J per bit. This is the floor that aestivation proposes to lower by waiting $\sim 10^{12}$ years (Sandberg et al., 2016).

An event horizon lowers it in the present epoch. By the generalized second law (GSL), the sum of black-hole entropy $S_{\text{BH}} = k_B c^3 A / 4G\hbar$ and exterior entropy is non-decreasing, and entropy carried across the horizon is accounted by the area increase (Bekenstein, 1973, 1974; Bousso, 2002). A computation platform orbiting the hole may therefore dump its waste entropy into the horizon rather than radiating it to the sky. The marginal energy cost of horizon disposal is set by the horizon temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_B} \simeq 6.2 \times 10^{-8} \left(\frac{M_\odot}{M} \right) \text{ K}, \quad (4)$$

(Hawking, 1974, 1975): $T_H \simeq 3 \times 10^{-12}$ K at $2 \times 10^4 M_\odot$, twelve orders of magnitude below the CMB. A bit dumped at the horizon costs $k_B T_H \ln 2 \sim 3 \times 10^{-35}$ J against 2.6×10^{-23} J for CMB radiation (Fig. 3). In practical terms the erasure cost ceases to be the binding constraint on computation; the budget shifts to transport of waste heat to the horizon (radiative or conductive tethering from the ISCO swarm inward), an engineering loss rather than a thermodynamic floor. The quantified gain from this substitution is the $\sim 10^{12}$ temperature ratio between the CMB and the horizon; the larger $\sim 10^{30}$ factor cited by Sandberg et al. (2016) assumes cooling to temperatures far below the horizon temperature used here and is not directly comparable. The substitution's decisive property is that it operates without waiting, and therefore without forfeiting the free energy that Bennett et al. (2019) showed dormant civilizations irretrievably lose. The related configuration of Opatrný et al. (2017), a shell absorbing CMB radiation as its hot reservoir and rejecting entropy to a central black hole, demonstrates the same thermodynamic asymmetry in a fully worked classical setting.

The claim should also be stated in its energy-conservation form, because that is where the mainstream objection lands. Curtis et al. (2026) state the standard position: Dyson-scale computation must reradiate nearly all of the energy it absorbs, so mid-infrared thermal emission is the robust technosignature of black-hole-hosted intelligence. For a horizon-sink architecture the premise fails, and it fails on energy grounds rather than entropy grounds: the waste carriers themselves cross the horizon, and their energy goes with the entropy. Writing f_{sink} for the fraction of waste *energy* delivered into the capture cone (a vocabulary adopted throughout this series), only the missed fraction $(1 - f_{\text{sink}})$ must be reradiated, at swarm temperature; the absorbed remainder raises M and thereby lowers T_H further, so the channel is weakly self-improving. The disagreement with Curtis et al. (2026) is thus quantitative and falsifiable: the predicted mid-infrared residue scales with $(1 - f_{\text{sink}})$ rather than with the full computational throughput.

The $k_B T_H \ln 2$ figure is the marginal cost of *disposal at the horizon*; it is not the delivered cost from an orbiting platform, and the gap between the two can be estimated rather than merely flagged. Consider radiative delivery, the least demanding channel. Efficient absorption requires a carrier wavelength $\lambda \lesssim r_g$, the geometric-optics condition for capture rather than diffraction around the hole (aiming is a separate and easier requirement, treated below). At $M = 2 \times 10^4 M_\odot$, $r_g \simeq 3 \times 10^7$ m, so the vacuum-gravitational floor on the carrier energy is $E_\gamma \gtrsim \hbar c / r_g \simeq 7 \times 10^{-33}$ J, corresponding to a frequency of ~ 10 Hz. That floor is unreachable through plasma: propagation requires $\nu > \nu_p = 8.98 \text{ kHz} \sqrt{n_e / \text{cm}^{-3}}$, and even the cluster's ambient $n_e = 0.23 \text{ cm}^{-3}$ gives $\nu_p \simeq 4.3 \text{ kHz}$, while any fed inner-flow environment ($n_e \sim 10^4\text{--}10^8 \text{ cm}^{-3}$ inside $\sim 10^2 r_g$) gives $\nu_p \simeq 1\text{--}90 \text{ MHz}$. The minimum propagating carrier energy is therefore set by the plasma cutoff, at $\sim 10^{-27}\text{--}10^{-26}$ J, three to six decades above $\hbar c / r_g$. Two consequences follow. First, single-bit-per-photon delivery no longer suffices: at the plasma-cutoff carrier energy the advantage over the CMB would erode toward $10^4\text{--}10^7$. What preserves the realized range is dense coding: a carrier may

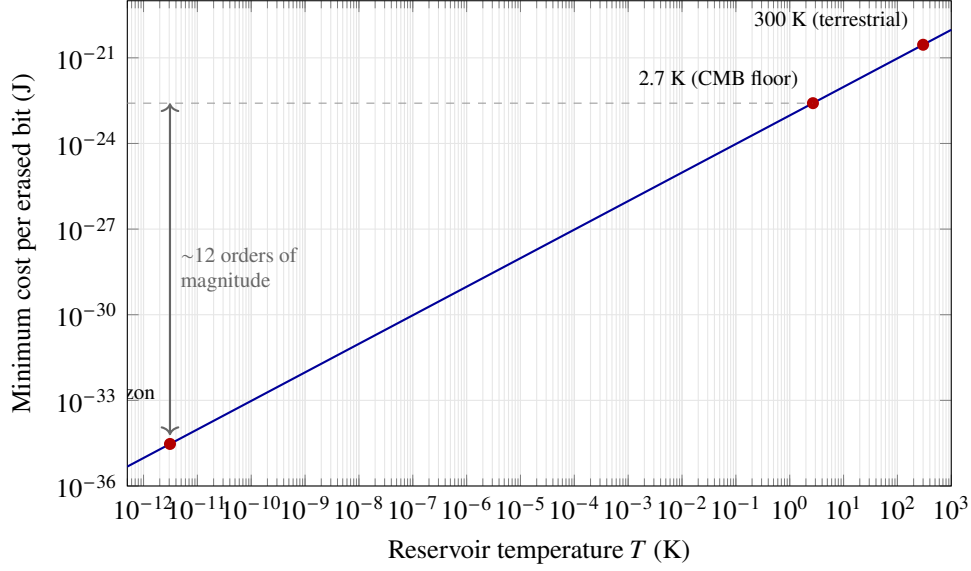


Figure 3: Landauer cost per erased bit, $k_B T \ln 2$, as a function of reservoir temperature. Dumping waste entropy across an IMBH horizon (marginal temperature T_H , Eq. 4) rather than radiating to the CMB lowers the erasure floor by $\sim 10^{12}$, in the present epoch, consistent with the generalized second law (Bekenstein, 1974). The aestivation hypothesis (Sandberg et al., 2016) sought a comparable-in-kind gain by waiting $\sim 10^{12}$ years for cosmological cooling, to still lower assumed reservoir temperatures.

bear many bits, up to the Bekenstein channel capacity (Bekenstein, 1981; Bousso, 2002), and coding at a modest fraction of capacity restores the delivered cost per bit to a factor of 10^6 – 10^9 below the CMB figure of 2.6×10^{-23} J. Second, matter carriers evade the cutoff entirely: cold mass dropped down the capture cone crosses the horizon with its entropy and energy aboard and no dispersion relation to satisfy, and serves as the fallback channel wherever the local plasma forbids cheap photons. Charging realistic engineering losses against these channels (pointing and capture inefficiency, error-correction overhead, the fraction of carriers scattered rather than absorbed) leaves a defensible realized gain of $\sim 10^6$ – 10^9 over the 2.7 K sky. Gravitational bookkeeping does not change this range: energy accounted at a platform of radius $r \gtrsim 10^2 r_g$ is redshifted at infinity by the platform’s lapse factor, a ≈ 1 per cent correction at these radii. The realized economy therefore sits well above the $k_B T_H \ln 2$ floor and well below the CMB figure, and we quote 10^6 – 10^9 rather than the floor-to-floor ratio of $\sim 10^{12}$ wherever the realized gain is meant. An improvement of six orders of magnitude over the 2.7 K sky already dominates the aestivation trade.

Delivery geometry can be derived rather than assumed. The photon-capture cross-section of a Schwarzschild hole is $\sigma = \pi b_{\text{crit}}^2 = 27\pi r_g^2$, with critical impact parameter $b_{\text{crit}} = 3\sqrt{3} r_g$; from a platform at $r = 10^2 r_g$ the capture target subtends a solid-angle fraction $\Omega_c/4\pi = 6.75 \times 10^{-4}$, so isotropic emission delivers less than a thousandth of its power to the horizon. Reaching $f_{\text{sink}} \rightarrow 1$ therefore requires a beaming gain $\gtrsim 1.5 \times 10^3$, which is optically trivial: the capture cone has half-angle $\theta_c \simeq 0.052$ rad, so diffraction demands only an aperture $D \gtrsim 23\lambda$, and the étendue (concentration) limit $1/\sin^2 \theta_c \approx 3.7 \times 10^2$ puts the optic area at a few hundred times the radiator area. (The $27\pi r_g^2$ figure is the Schwarzschild value; the near-extremal prograde equatorial cross-section is somewhat smaller and anisotropic, an $O(1)$ correction.) The binding term in the delivery budget is re-interception within the swarm rather than aiming or aperture: beamed flux must traverse the swarm itself, and the intercepted fraction scales with the swarm covering fraction (Swanson, 2026).

4.3 Information storage: the Bekenstein–Hawking archive

The Bekenstein bound limits the information content of any bounded system (Bekenstein, 1981; Casini, 2008); black holes saturate it, with

$$\frac{S_{\text{BH}}}{k_B \ln 2} \simeq 1.5 \times 10^{77} \left(\frac{M}{M_\odot} \right)^2 \text{ bits}, \quad (5)$$

i.e. $\sim 6 \times 10^{85}$ bits at $2 \times 10^4 M_\odot$ (Bekenstein, 1973; Hawking, 1975; Lloyd, 2000). That coefficient is the Schwarzschild value; since the architecture ends near-extremal, we quote one self-consistent pair: at $a_\star = 0.998$ the Kerr horizon area per unit M^2 is $0.53 \times$ Schwarzschild, so a $2 \times 10^4 M_\odot$ hole at that spin stores $\approx 3 \times 10^{85}$ bits, and the factor-2.2 mass growth of the spin-up path (Section 4.1) more than recovers the difference through the M^2 scaling. Information crossing the horizon is scrambled but, by unitarity, not destroyed; retrieval timescales via Hawking radiation are, however, of order the evaporation time, $\sim 10^{67} (M/M_\odot)^3$ yr (Page, 1976): the archive is functionally write-only. Black holes likewise saturate the Margolus–Levitin bound on processing rate, $2E/\pi\hbar$ operations per second (Margolus and Levitin, 1998; Lloyd, 2000). We treat these saturation properties as the formal statement that no material technology can out-store or out-compute the object the civilization is already orbiting.

4.4 Time dilation as an engineering parameter

A platform on a circular equatorial geodesic at radius r around a Kerr hole runs slow relative to infinity by

$$\frac{d\tau}{dt} = \frac{\sqrt{1 - 3r_g/r + 2a_\star(r_g/r)^{3/2}}}{1 + a_\star(r_g/r)^{3/2}}, \quad (6)$$

(Bardeen et al., 1972): $d\tau/dt = \sqrt{1/2} \simeq 0.707$ at the Schwarzschild ISCO, falling to $\simeq 0.093$ (a factor ~ 11) at the prograde ISCO for $a_\star = 0.998$ (Fig. 4). Larger factors require powered non-geodesic hovering at diverging thrust cost and are not usable by stable platforms. The $\sim 11:1$ ceiling is set by the Thorne (1974) spin-equilibrium limit $a_\star = 0.998$ rather than by orbital stability: the ISCO proper-time ratio falls toward zero as $a_\star \rightarrow 1$, and it is the accretion physics capping the spin that caps the dilation. A tiered swarm can therefore place archival and slow-integration processes deep (subjectively skipping across external time) and interactive processes high, with the choice of orbit acting as a clock-rate dial, a resource with no terrestrial analogue, though we note its strategic value (e.g. outwaiting external change) trades directly against responsiveness. The dial is two-way: viewed from outside, a deep tier both runs and receives $11\times$ slower in external time, so deep placement taxes bandwidth to, and coherence with, the shallow tier; and uplink photons arrive blueshifted by the same factor, each depositing $\sim 11\times$ its emitted energy at the deep receiver.

4.5 Inward versus outward migration

The expansionist default implicitly maximizes resource acquisition; P1 civilizations maximize computation, and the two diverge. Expansion multiplies harvested mass polynomially in time ($\propto t^3$ for spherical growth at fixed speed) while imposing light-lag decoherence on any shared computation: a civilization spread over 10^3 light-years cannot function as one computer on subjective timescales shorter than millennia. Concentration, by contrast, multiplies the yield per unit mass by the factors of Section 4.1 (up to ~ 600 over Dyson harvesting without star lifting), lowers the realized erasure cost by 10^6 – 10^9 (Section 4.2), and keeps the entire system within light-seconds of itself. We must be careful about what this establishes.

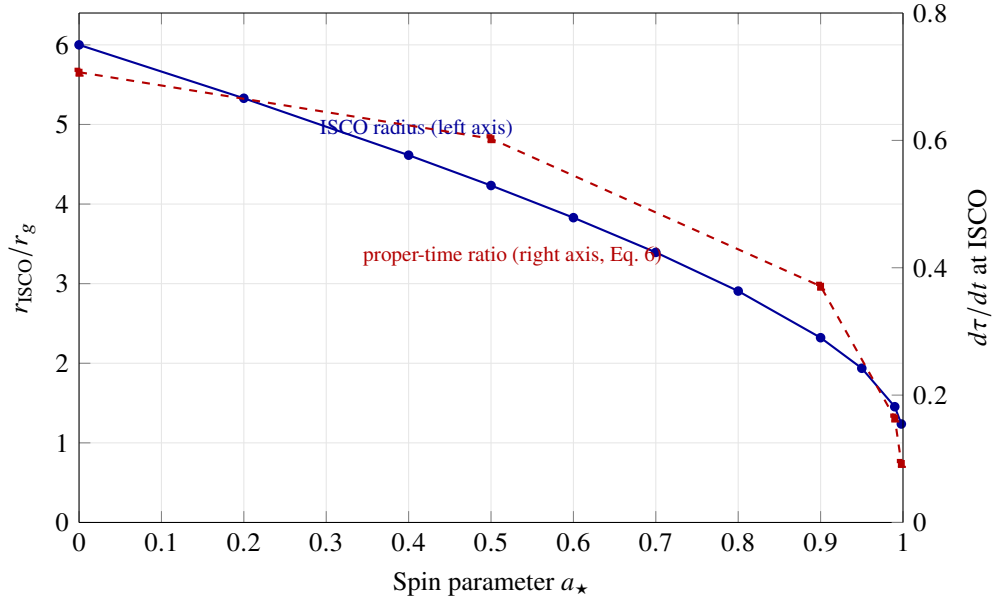


Figure 4: Prograde ISCO radius (left axis) and proper-time ratio for a circular geodesic at the ISCO (right axis) as functions of spin (Bardeen et al., 1972). Spin-up from $a_* = 0$ to the Thorne limit moves the working orbit from $6 r_g$ to $1.24 r_g$ and deepens the available stable time dilation from 1.41:1 to $\sim 11:1$.

An undiscounted maximizer of total integrated computation over unbounded time is served by expansion despite the per-unit-mass penalty, because controlled mass grows as t^3 and per-unit efficiency gains are bounded. The gradient points inward only under at least one of three auxiliary assumptions, which we adopt as part of P1: (i) temporal discounting, so that near-term computation outweighs the polynomial tail; (ii) a requirement that the computation be unified and latency-bound, so that light-lag decoherence caps the useful radius; or (iii) bounded planning horizons, within which the concentration factors dominate the t^3 growth. Under any of these, beamed-sail transport makes a cluster-hosted IMBH reachable at negligible marginal cost (Lubin, 2016; Armstrong and Sandberg, 2013) and concentration wins. We stress the modesty of the claim actually needed: P1 requires only that some civilizations satisfy one of (i)–(iii) and follow the gradient. Expansionist and stay-at-home civilizations may coexist with migrators; the Fermi-relevant point is that the oldest and most capable optimizers, the very civilizations whose absence the paradox finds most puzzling (Hanson et al., 2021), are the ones the gradient captures.

4.6 One hole rather than many

The comparison set for P2 must include a rival platform that the falsification program itself puts on the table: the dark remnant swarm of hypothesis H_1 (Section 7). Under H_1 the centre of ω Cen contains an extended $2\text{--}3 \times 10^5 M_\odot$ component of stellar-mass black holes (Bañares-Hernández et al., 2025), twelve times the mass of the fiducial $2 \times 10^4 M_\odot$ IMBH. Because $L_{\text{Edd}} \propto M$, the swarm’s aggregate Eddington power is $\sim 3.2 \times 10^{36}$ W against the IMBH’s 2.5×10^{35} W: on raw power the “falsifying” configuration is the better platform by a factor of ~ 13 , and a selection theory that ignored this would be assuming its conclusion. Table 2 prices the two configurations against the P2 axes.

The IMBH wins decisively on three of the four P2 axes and loses on the fourth. Storage scales as $\sum M_i^2$, so fragmenting the central mass into 2.5×10^4 holes of $10 M_\odot$ costs a factor of 160 even against the twelvefold mass advantage. Sink depth scales as $T_H \propto M^{-1}$, a 2000 $\times$ edge for the IMBH, and the edge

Table 2: One IMBH versus a same-order swarm of stellar-mass holes, priced on the P2 axes. The swarm column takes 2.5×10^4 holes of $10 M_\odot$ (total $2.5 \times 10^5 M_\odot$, the upper end of the Bañares-Hernández et al. 2025 extended component); the IMBH column takes the fiducial $2 \times 10^4 M_\odot$. Storage and tidal entries are quoted as ratios (arb. units).

Figure of merit	One $2 \times 10^4 M_\odot$ hole	2.5×10^4 holes of $10 M_\odot$	Advantage
Aggregate L_{Edd}	2.5×10^{35} W	3.2×10^{36} W	swarm, 13×
Storage, $\sum S_i \propto \sum M_i^2$	4×10^8	2.5×10^6	IMBH, 160×
Sink depth, $T_H \propto M^{-1}$	3.1×10^{-12} K	6.2×10^{-9} K	IMBH, 2000×
ISCO tidal field, $\propto M^{-2}$	1	4×10^6	IMBH, 4×10^6
Latency across working system	ms (one orbit)	μ s per hole; pc-scale between holes	IMBH

survives the delivery-level accounting of Section 4.2: the realized erasure economy tracks the delivered carrier cost rather than T_H directly, and the vacuum carrier floor hc/r_g is 1.3×10^{-29} J for a $10 M_\odot$ hole against 6.7×10^{-33} J for the IMBH, the same $\sim 2 \times 10^3$ ratio. The ISCO tidal field, $\propto M^{-2}$ at fixed r/r_g , is 4×10^6 times harsher at $10 M_\odot$, excluding all but dust-scale hardware from the deep orbits where the dilation and efficiency gains live. And a computation distributed over the swarm pays parsec-scale light lag between nodes, against milliseconds across a single working orbit, failing the latency-bound condition of Section 4.5 by construction. A computation-maximizer offered both configurations takes the single hole; consolidation is worth a large premium under the same objective.

This derivation does real work in the falsification program. T3 (Table 5) treats a LISA-resolved remnant swarm as a decisive falsifier, and this subsection is why: the swarm loses on storage, sink depth, and tides by two to six orders of magnitude, so a maximizer with gigayears of run of the cluster would not have left the central mass in that configuration. T3 is thereby a derived kill switch rather than a stipulated one.

5 Omega Centauri as the optimal observational target

5.1 The cluster

ω Centauri is the most massive Galactic globular cluster ($M_{\text{cl}} \approx 4 \times 10^6 M_\odot$, $\sim 10^7$ stars; Harris 1996), at a kinematic distance of 5.49 ± 0.06 kpc (Häberle et al., 2025), in $\sim 2\sigma$ tension with the Gaia EDR3 parallax (5.24 ± 0.11 kpc; Soltis et al. 2021), whose error budget is dominated by the parallax zero-point systematics discussed by those authors, and consistent with the combined catalogue value (5.43 ± 0.05 kpc; Baumgardt and Vasiliev 2021); nothing in this paper depends on the difference. It is anomalous among globular clusters in nearly every respect that matters here: it hosts multiple stellar populations spanning a broad metallicity range (Johnson and Pilachowski, 2010), a mean stellar age of 12.08 ± 0.01 Gyr with an intrinsic spread of 0.75 Gyr (Clontz et al., 2024), significant rotation, and an associated tidal stream (Ibata et al., 2019); the consensus interpretation is that it is the surviving nucleus of a dwarf galaxy accreted and stripped by the Milky Way (Hilker and Richtler, 2000; Bekki and Tsujimoto, 2019; Souza et al., 2026). Table 3 summarizes the parameters used in this paper.

For the MTH selection criteria of Section 3.1, the stripped-nucleus origin matters twice over. First, nuclear star clusters are the environments most likely to form and retain IMBHs (Greene et al., 2020). Second, the age structure implies that any technological lineage arising anywhere in the progenitor dwarf (or in the early Milky Way) has had $\gtrsim 8$ Gyr to locate, reach, and develop the system: at beamed-sail transit speeds of 0.1–0.2 c (Lubin, 2016), crossing the Galaxy takes $\lesssim 10^6$ years, roughly eight thousand times

Table 3: Adopted parameters for ω Centauri (NGC 5139).

Quantity	Value	Source
Distance	5.49 ± 0.06 kpc	Häberle et al. (2025)
Cluster mass	$\approx 4 \times 10^6 M_\odot$	Harris (1996); Baumgardt and Vasiliev (2021)
Stellar count	$\sim 10^7$	Harris (1996)
Mean stellar age	12.08 ± 0.01 Gyr (spread 0.75 Gyr)	Clontz et al. (2024)
Origin	stripped dwarf-galaxy nucleus	Hilker and Richtler (2000); Ibata et al. (2019); Souza et al. (2026)
Central density	$\sim 3 \times 10^3 M_\odot \text{pc}^{-3}$	Baumgardt and Vasiliev (2021); cf. $\sim 10^3$ in Pryor and Meylan (1993)
Declination	-47.5°	Harris (1996)

shorter than the available window.

5.2 The IMBH candidate and the mass tension

Evidence for a central dark mass in ω Cen has accumulated for two decades, from integrated kinematics (Noyola et al., 2008) through proper-motion modelling that initially yielded only upper limits at the $\sim 10^4 M_\odot$ level (van der Marel and Anderson, 2010), with stellar-mass black-hole subsystems long advanced as the alternative explanation (Zocchi et al., 2019; Breen and Heggie, 2013) and a systematic N -body survey concluding that no Galactic globular cluster, ω Cen included, required an IMBH at all (Baumgardt et al., 2019). The situation changed qualitatively when Häberle et al. (2024b), using a proper-motion catalogue of 1.4 million stars built from over 500 HST epochs (Häberle et al., 2024a), identified seven stars within 3 arcsec of the cluster centre moving faster than the local escape velocity. Their velocities alone require a point mass $\geq 8,200 M_\odot$; including acceleration limits on the same stars raises the lower bound to $21,100 M_\odot$ at 99 per cent confidence (main text of Häberle et al. 2024b).¹ Independent N -body modelling finds that an IMBH grown in situ to $\sim 5 \times 10^4 M_\odot$ reproduces the present-day cluster structure and fast-star population (González Prieto et al., 2025).

Against this stands Bañares-Hernández et al. (2025): a joint analysis of stellar kinematics and timing accelerations of the cluster’s millisecond pulsars that places a 3σ upper limit of $\sim 6 \times 10^3 M_\odot$ on any central point mass, favouring instead an extended central dark component of $\approx 2\text{--}3 \times 10^5 M_\odot$, naturally interpreted as a centrally concentrated cluster of stellar remnants. The most recent joint MeerKAT–Parkes timing analysis, by contrast, finds its measurements insensitive to an IMBH in the $10^3\text{--}10^4 M_\odot$ range and places a 90 per cent upper limit of $< 10^5 M_\odot$ on the central mass (Colom i Bernadich et al., 2026), a bound compatible with both camps. The two leading results are formally inconsistent under each side’s stated modelling assumptions, and the disagreement is unresolved at the time of writing (Table 4). We take no side; for this paper, the tension is what makes the system valuable as a target. The MTH requires a genuine IMBH (P2), so the Bañares-Hernández et al. (2025) scenario, if confirmed, falsifies the ω Cen application outright (one of several clean kill switches catalogued in Section 7). LISA can settle the question definitively, contingent on a suitable source being in-band during the mission: a single IMBH undergoing a compact-object inspiral yields resolvable extreme- or intermediate-mass-ratio-inspiral (EMRI/IMRI) signals with percent-level mass and spin measurement, whereas a remnant swarm yields a qualitatively different gravitational-wave signature (Amaro-Seoane, 2018; Babak et al., 2017; Colpi et al., 2024).

¹The abstract of Häberle et al. (2024b) quotes only the velocity-only $8,200 M_\odot$ limit; the $21,100 M_\odot$ figure appears in the main text. A Nature *Author Correction* (doi:10.1038/s41586-024-08017-4) fixes an equation typo and a right-ascension offset; the results are unaffected.

Table 4: Current constraints on the central dark mass of ω Cen. The kinematic lower bounds and the pulsar-timing upper bound are mutually inconsistent under stated assumptions; resolution is expected from LISA EMRI observations and continued astrometry.

Constraint	Value	Method / source
Lower bound (velocities only)	$\geq 8,200 M_{\odot}$	7 fast stars, HST proper motions (Häberle et al., 2024b)
Lower bound (with accelerations)	$\geq 21,100 M_{\odot}$ (99%)	same stars, acceleration limits (Häberle et al., 2024b)
N -body growth models	$\sim 4.7\text{--}5.1 \times 10^4 M_{\odot}$	cluster-evolution simulations (González Prieto et al., 2025)
Upper bound (point mass)	$< 6 \times 10^3 M_{\odot}$ (3σ)	stellar kinematics + MSP timing (Bañares-Hernández et al., 2025)
Favoured alternative	extended $2\text{--}3 \times 10^5 M_{\odot}$	same analysis (Bañares-Hernández et al., 2025)
Timing (joint MeerKAT–Parkes)	insensitive at $10^3\text{--}10^4 M_{\odot}$; $< 10^5 M_{\odot}$ (90%)	MSP timing (Colomi Bernadich et al., 2026)
Historical upper limits	$\lesssim 10^4 M_{\odot}$ (model-dep.)	HST proper motions (van der Marel and Anderson, 2010); see also Zocchi et al. (2019)

The contingency deserves emphasis, because the per-cluster IMRI rate for a $\sim 10^4 M_{\odot}$ hole is uncertain by orders of magnitude, and the plausible outcome of a 4–10 yr mission is that no inspiral is caught in-band, in which case the gravitational-wave branch of Section 7 returns “inconclusive” rather than a verdict (Fig. 6, silence branch). Two mitigations exist. The dense fast-star environment resolved by Häberle et al. (2024b) raises the prior on close compact companions relative to a field IMBH, though it does not remove the rate uncertainty. More important, an inspiral is not the only LISA channel: Stokov et al. (2022) show that a Galactic-cluster IMBH can reveal itself through Doppler modulation of the gravitational-wave signal of an orbiting stellar-mass binary, a continuous signature that requires no merger event during the mission. This channel constrains the mass, though not the spin, so the decisive T5 measurement remains inspiral-contingent.

One element of the predicted dark population has now been observed directly. Whitaker et al. (2026) report the astrometric detection of a $4.46^{+1.22}_{-1.01} M_{\odot}$ black hole with reported orbital elements $P = 94$ yr, $e = 0.72$, $a = 31$ AU and a $\sim 0.8 M_{\odot}$ main-sequence companion,² from HST astrometry spanning 2002–2023 combined with JWST imaging: the first confirmed stellar-mass black hole in ω Cen, and the first astrometric detection of one in any globular cluster. The detection confirms that the cluster forms and retains stellar-mass black holes, the population of which both sides of the mass tension require ($\sim 10^4$ such remnants in the extended-component scenario; a formation reservoir in the IMBH-growth scenario), and it anchors the low-mass end of the central mass budget with an object whose parameters are measured rather than modelled.

5.3 Electromagnetic silence

The centre of ω Cen is extraordinarily dark, whatever occupies it. The deepest radio observation of any globular cluster (~ 170 h with ATCA at 7.25 GHz, reaching an rms of $1.1 \mu\text{Jy beam}^{-1}$) detects nothing at

²We note without resolution that the reported elements are in mild internal tension: Kepler’s third law with $a = 31$ AU and $M_{\text{tot}} \simeq 5.3 M_{\odot}$ gives $P \simeq 75$ yr; the reported period exceeds this Kepler-implied value by ~ 25 per cent; reconciliation presumably lies within the quoted uncertainties on a and the masses.

any proposed cluster centre, bounding the accretion efficiency of a putative IMBH at $\epsilon \lesssim 4 \times 10^{-3}$ of the Bondi rate (3σ ; model-dependent, with ≈ 1 dex of systematic from the fundamental-plane conversion) (Mahida et al., 2026), the deepest point yet in the radio-limit framework established for globular clusters by the MAVERIC survey (Tremou et al., 2018). JWST NIRCам and MIRI photometry of the central field likewise finds no source with the spectral energy distribution of an accreting IMBH, with limits most constraining at the low-mass end of the allowed range (Chen et al., 2025). The X-ray channel, the standard probe of quiescent accretion, agrees: a ~ 291 ks Chandra exposure detects no central point source, limiting the unabsorbed luminosity to $L_X(0.5\text{--}7.0\text{ keV}) \lesssim 1.6 \times 10^{30} \text{ erg s}^{-1}$, an Eddington ratio of $\lesssim 10^{-12}$ for a $10^4 M_\odot$ hole (Haggard et al., 2013). To our knowledge, no dedicated technosignature search of ω Cen has been conducted at any wavelength, although the commensal opportunity is unusually rich: ω Cen is a flagship MeerKAT pulsar-timing target, so every timing session is a potential commensal SETI epoch on the cluster for Breakthrough Listen’s MeerKAT backend; the first dedicated globular-cluster SETI survey (Huang et al., 2026b, five northern clusters with FAST; ω Cen is inaccessible from FAST’s latitude) demonstrates both the feasibility and the current emptiness of this niche, and a Gaia-based phase-space anomaly metric over 79 Galactic globular clusters, in which NGC 5139 ranks among the dynamically most complex systems proposed for technosignature-oriented follow-up (Huang et al., 2026a), provides independent validation for treating globular-cluster anomalies as search targets.

The logic matters here: the MTH predicts electromagnetic silence (P3–P4), but silence is equally consistent with, and more parsimoniously explained by, a quiescent, gas-starved compact object in a relaxed, gas-poor old cluster, which we adopt as the null hypothesis H_0 throughout. Silence cannot confirm the MTH; only the discriminating residues of P4 can separate the hypotheses (Section 7).

5.4 The case against Sagittarius A*

A natural objection: if bigger is better, the Galactic Centre hosts a $4.3 \times 10^6 M_\odot$ hole (GRAVITY Collaboration, 2022; Event Horizon Telescope Collaboration, 2022) with abundant fuel. Three of the four selection criteria of Section 3.1 disfavour it; the first is thermodynamic, the second and third strategic (per the register of claim 1, Section 1.1). First, the storage and erasure advantages saturate at IMBH scale: nothing a maximizer needs scales usefully beyond $\sim 10^{4-5} M_\odot$, while hazards do. Second, the Galactic Centre is the most dynamically violent environment in the Galaxy (ongoing star formation, supernovae, magnetar flares, stochastic accretion flares, and a dense population of perturbers), the opposite of a stable substrate for Gyr-scale infrastructure; ω Cen’s relaxed core offers the same physics in a far quieter dynamical environment. Third, the claim structure matters, in the era-relative form of criterion 3: the Galactic Centre’s gas and stars are the future fuel of natural astrophysics and of any fusion-era competitor, whereas a globular cluster’s low-mass reservoir is uncontested by fusion-era economics and contested only among computation-maximizing lineages, the population P1 already conditions on; the premium therefore goes to early arrival rather than to contest avoidance. The MTH thus predicts that the preferred class is IMBHs in dense old clusters rather than supermassive holes, a distinctive and testable preference, since it directs attention at objects mainstream SETI has never examined.

6 A staged exploitation architecture

Epistemic status: This section is speculative civilizational engineering. It is included not as prediction but because the falsification program of Section 7 requires a concrete model of what would be observable at each stage. Each stage uses only physics established in Section 4 plus engineering extrapolations that are individually defended in the cited literature; the conjunction is unargued-for and carries the usual burden of compounded speculation.

We model the migration as five phases (Fig. 5), parameterized for a civilization departing a planetary system ~ 5 kpc from ω Cen.

Phase 1 — Scouts. Gram-scale beamed-sail probes at $0.1\text{--}0.2\,c$ (Lubin, 2016), with photon-braking and magnetic-sail deceleration into the cluster halo; transit $\sim 10^5$ yr. Objectives: resolve the IMBH-versus-swarm question in situ, map the rocky-body inventory, and emplace a deceleration relay for following waves. *External signature: none detectable at interstellar range.*

Phase 2 — Self-replicating industry. Kilogram-scale seed factories bootstrap exponentially from asteroidal material, following the NASA self-replicating-systems concept (Freitas and Gilbreath, 1982); decades to centuries to 10^{3+} factory units and a relay laser. *Signature: waste heat of order early-industrial — far below detectability.*

Phase 3 — Compute swarm. The main payload (substrate-independent minds, by this stage the default cargo; cf. Smart 2012) arrives and assembles a free-flying compute swarm. While the hole is being fed, the swarm’s inner working boundary sits at $\gtrsim 10^2\,r_g$ ($\sim 3 \times 10^6$ km at $2 \times 10^4\,M_\odot$): the companion infrastructure analysis (Swanson, 2026) finds that accretion-flow erosion excludes hardware inside $\sim 30\,r_g$ of a fueled hole, and differential Lense–Thirring precession already excludes rigid formations at $10^2\,r_g$, so the swarm must be free-flying there in any case. Controlled star-lifting of brown dwarfs establishes a thin sub-Eddington disk; BZ extraction begins at low spin. Tiered orbits allocate clock rate (Section 4.4), though for an operating (fueled) system the stable dilation on offer at the $10^2\,r_g$ working boundary is modest, $d\tau/dt \simeq 0.985$; the deep-orbit clock rates of Fig. 4 become available only to dormant or archival elements around an unfueled hole. *Signature: faint, soft accretion luminosity — the one phase in which the system would be conventionally observable; at ω Cen, current limits already exclude an active Phase 3 fed above $\sim 4 \times 10^{-3}$ of the Bondi rate (Mahida et al., 2026) or radiating above $\sim 10^{-12}$ of Eddington (Haggard et al., 2013; Chen et al., 2025).*

Phase 4 — Spin-up. Sustained prograde feeding ($\sim 10^{4-5}$ stars over $\sim 3.5 \times 10^7$ yr, 0.79 Salpeter e-foldings to reach Thorne’s radiation-limited $a_\star = 0.998$ ($M_f/M_i \approx 2.2$) (Bardeen, 1970; Thorne, 1974; Swanson, 2026); Section 4.1 drives $a_\star \rightarrow 0.9\text{--}0.998$; the ISCO migrates inward by a factor ~ 5 and jet efficiency crosses unity in MAD episodes; the fueled swarm holds its $\sim 10^2\,r_g$ working boundary throughout (Swanson, 2026). *Signature: episodic jet activity if imperfectly managed; high spin as a permanent record. Secular cluster-core depletion is a weaker observable than earlier drafts implied: the delivered $(3\text{--}4) \times 10^4\,M_\odot$ is ~ 24 per cent of the mass inside $2\,\text{pc}$ but only ~ 1 per cent of the mass inside the 10-pc reservoir the fuel census requires, the removed stellar mass reappears as central point mass so the total enclosed profile is preserved, and any deficit refills on the $\sim 4\text{-Gyr}$ core relaxation time (Harris, 1996; Swanson, 2026).* We flag the strongest hidden premise of the whole architecture here: Phases 3–5 presuppose institutional or goal stability over 10^8 yr, longer than mammalian evolutionary history, and stellar feeding requires active orbit-shaping of individual stars with non-trivial Δv budgets. We know of no principled argument that either is achievable; the architecture is offered conditional on both.

The Δv can at least be priced to order of magnitude, and the pricing rules out two of the three options a reader might imagine. Delivering a star inward from a near-circular cluster orbit requires removing nearly all of its orbital angular momentum; the per-star impulse is of order the local orbital speed, tens of km s^{-1} across the central $1\text{--}10\,\text{pc}$ and growing with the radius of the $\sim 10\text{-pc}$ reservoir that the fuel census of Section 3.1 requires. The energy is not the obstacle (binding energy is released on the way down); the angular momentum is.

Natural refilling does not supply the rate. Phase 4 needs 10^{4-5} stars over $\sim 3.5 \times 10^7$ yr, i.e. $\sim 3 \times 10^{-4}\text{--}3 \times 10^{-3}$ stars yr^{-1} . Natural loss-cone refilling, which resonant relaxation drives near the influence radius (Rauch and Tremaine, 1996), delivers stars to the hole at the tidal-disruption rate, $\sim 10^{-8}\text{--}10^{-7}\,\text{yr}^{-1}$ for IMBHs in evolved clusters and $\sim 5 \times 10^{-8}\,\text{yr}^{-1}$ in the ω Cen models of González Prieto et al. (2025)

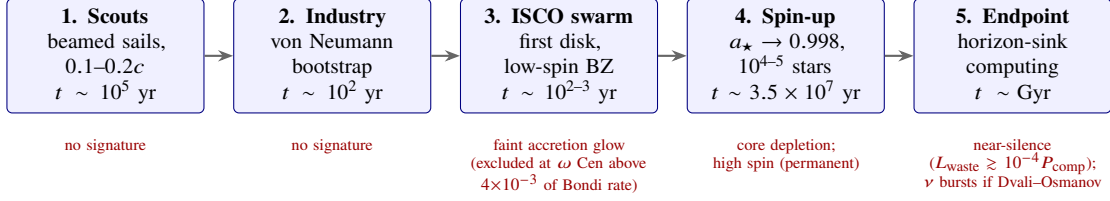


Figure 5: The five-phase exploitation architecture (speculative; see epistemic-status box) and the external signature of each phase. Only Phase 3 is conventionally luminous; the permanent observables are dynamical (spin, core depletion), which is why gravitational-wave spin measurement is the decisive test.

specifically (Swanson, 2026). The shortfall is four to five orders of magnitude. The natural channel is not a slower alternative to engineered delivery; it is not a fuel supply for this architecture at all, and essentially the whole Phase-4 budget must be actuated.

Beamed-momentum tugs are excluded by the waste-heat ceiling. Imparting $\Delta v \sim 10^2 \text{ km s}^{-1}$ to the $\sim 3 \times 10^4 M_{\odot}$ that Phase 4 consumes costs $\sim 3 \times 10^{44} \text{ J}$ of kinetic work, $\sim 7 \times 10^2 L_{\odot}$ averaged over $3.5 \times 10^7 \text{ yr}$, which fits inside the concealment budget with room to spare. Paying that momentum in photons does not: a photon tug needs $E = \Delta p c \simeq 1.8 \times 10^{48} \text{ J}$, or $\sim 4.2 \times 10^6 L_{\odot}$ sustained over the $3.5 \times 10^7 \text{ yr}$ episode, $5.6\times$ the ambient fuel ceiling and two and a half to three and a half orders of magnitude above the mid-infrared limits that already bound an operating system at ω Cen (Swanson, 2026). The photons are absorbed by the star being moved and reradiated at stellar temperature across the cluster, so there is nowhere to put them. Gravitational-assist sequencing among cluster stars, which uses other stars as momentum reservoirs and pays only actuator overhead, is the one scheme that survives its own thermal signature.

Delivery geometry, not merely delivery rate, is what the spin fossil records. Loss-cone stars arrive isotropically, and isotropic feeding random-walks a hole’s spin rather than raising it (Section 7; Swanson 2026). Coherent prograde delivery is therefore not an efficiency improvement on the natural channel but the only channel that produces the observable this paper treats as its decisive test. A full account (which stars, in what order, at what actuator cost, in what orbital orientation) has not been produced and remains the largest quantitative gap in this section.

Phase 5 — Endpoint. Near-extremal spin; waste entropy disposed across the horizon (Section 4.2); residual stellar population dispersed or consumed. During fueled operation the swarm computes at the $\gtrsim 10^2 r_g$ working boundary ($d\tau/dt \simeq 0.985$); if the system instead enters a dormant, unfueled archival mode, elements can descend to the near-ISCO orbits of Fig. 4 and bank the full $\sim 11:1$ stable dilation. The system is nearly, but not perfectly, thermodynamically silent: transport losses in delivering waste entropy to the horizon impose a waste-heat floor $L_{\text{waste}} \gtrsim 10^{-4} P_{\text{comp}}$ (Swanson, 2026), so current JWST mid-infrared limits already bound the computational throughput of any operating Phase-5 system at ω Cen to $P_{\text{comp}} \lesssim 10^{3-4} L_{\odot}$; there is no radio leakage, and the horizon temperature is in the picokelvin range (Eq. 4). The sole conjectured emission channel is burst-mode: if intractable computations are delegated to manufactured micro black holes per Dvali and Osmanov (2023), their evaporation yields high-energy neutrino and gamma-ray transients. We note for completeness that the formation of micro black holes from focused radiation may be blocked by Schwinger-limit pair production (Álvarez-Domínguez et al., 2024; Breit and Wheeler, 1934; Schwinger, 1951); this conclusion is contested (Loeb, 2024), and the MTH does not depend on the channel — it affects only the strength of prediction T6 below.

7 Falsification framework and observational program

We define the competing hypotheses explicitly, then the tests.

H_0 (**gas starvation; null**). The central object, whether IMBH or remnant swarm, is electromagnetically silent for the ordinary reason: a relaxed, gas-poor, 12-Gyr-old cluster supplies nothing to accrete. Favoured by parsimony and adopted as default.

H_1 (**dark remnant swarm**). No IMBH exists; the central mass is an extended cluster of stellar remnants (Bañares-Hernández et al., 2025; Zocchi et al., 2019; Breen and Hoggie, 2013). Falsifies the ω Cen application of the MTH.

H_2 (**MTH**). An IMBH exists and is, or has been, the substrate of a migrated civilization (Section 6).

H_0 and H_2 predict identical electromagnetic appearances today (by design, since P3–P4 entail silence), so electromagnetic non-detections cannot distinguish them. They diverge on dynamical and high-energy observables. Table 5 states six tests with thresholds, instruments, and timelines; Fig. 6 arranges the decisive subset as a decision tree.

Three features of this matrix deserve emphasis. First, it contains genuine kill switches: T3 falsifies the ω Cen application outright, and T5 falsifies the only exploitation model we have offered; T4 is a demotion rather than a kill, removing ω Cen from the top of the P2 selection ranking while leaving a still-substantial power budget in place. The MTH cannot survive a confirmed remnant swarm or a non-spinning IMBH, except as vacuous speculation about other systems. Second, the LISA spin measurement is among the most informative numbers obtainable this generation (if an in-band inspiral delivers it; Section 5.2), though both the natural-spin null and the expected yield must be stated carefully. Natural formation channels span the spin range (Reynolds, 2021): runaway stellar collisions build IMBHs near $10^4 M_\odot$ without requiring rapid rotation of the remnant (Fujii et al., 2024), and incoherent minor-merger growth random-walks the spin toward a broad distribution centred near $a_\star \approx 0.2\text{--}0.5$ (Reynolds, 2021), so the T5 trigger $a_\star < 0.1$ occupies a small tail of the natural prior and the most probable natural outcome lands in the uninformative middle branch of Fig. 6. A measured $a_\star < 0.1$ still fires T5 cleanly; the hedge concerns how likely nature is to deliver a verdict, not what the verdict would mean. The paper’s own account of ω Cen’s origin (Section 5) is, however, a gas-rich dwarf-galaxy nucleus, and an IMBH grown there by coherent disk accretion generically reaches high spin naturally; because Blandford–Znajek spindown at ambient fueling operates on $\sim 4 \times 10^{11}$ yr timescales (Swanson, 2026), such natal spin survives to the present. A measured $a_\star \gtrsim 0.9$ is therefore supporting evidence rather than a diagnostic residue on its own: the discriminating observation is the conjunction of high spin, electromagnetic silence far below the ambient Bondi supply, and the regulation signature of T1 (P4a–b together). EMRI parameter estimation at LISA delivers spin to better than percent precision for favourable systems (Babak et al., 2017; Amaro-Seoane, 2018). Third, the program is cheap: T1–T2 are archival or piggyback analyses; T6 requires only a pointed monitoring program on an instrument already built for the southern sky (Adrián-Martínez et al., 2016; The KM3Net Collaboration, 2025), following the high-energy SETI logic of Lacki and DiKerby (2025) and the pipeline practices of standard point-source searches (Aartsen et al., 2020); the partially built ARCA array has already demonstrated discovery-class capability with the $\sim 220\text{-PeV}$ event KM3-230213A, the most energetic neutrino yet observed (The KM3Net Collaboration, 2025); and T3–T5 are free by-products of LISA’s planned EMRI science (Colpi et al., 2024).

The T6 burst criterion must clear the atmospheric-neutrino background, and an order-of-magnitude estimate shows it does under the stated cuts. Full-detector ARCA is expected to collect of order 10^3 up-going atmospheric muon-neutrino events per year above 10 TeV over the full up-going sky (effective

Table 5: Falsification matrix for the MTH at ω Cen. “Kills” indicates what a positive result rules out; tests marked $\dagger$ are decisive against the hypothesis as a whole rather than against one phase. LISA timeline assumes launch in the mid-2030s (Colpi et al., 2024); T3–T5 are additionally contingent on a compact-object inspiral being in-band during the mission (Section 5.2), with the Doppler-modulation channel of Stokov et al. (2022) providing a mass-only fallback.

#	Observation	Threshold	Instrument	Timeline	Kills
T1	Accretion luminosity from core (limit-tightening)	Current best $L_{\text{acc}} \approx 2.5 \times 10^{26}$ W ($\approx 10^{-9} L_{\text{Edd}}$) (Chen et al., 2025); near-term target one further decade, $\lesssim 10^{26}$ W. Supplementary trigger: sustained (> 1 yr) brightening to $L_{\text{acc}} > 10^{29}$ W	JWST MIRI; ATCA/MeerKAT; Chandra	now–2030	Each deepening null lowers the ceiling on managed accretion throughput; a brightening detection kills Phase 5 (managed environment). Bounds the accretion SED itself, where T2 bounds reradiated waste heat (Haggard et al., 2013; Mahida et al., 2026)
T2	Mid-IR waste-heat excess	$> 3\sigma$ over stellar-population model, 10–25.5 μm (MIRI imaging limit); tested against the horizon-sink floor $L_{\text{waste}} \gtrsim 10^{-4} P_{\text{comp}}$ (Swanson, 2026)	JWST MIRI	now–2030	Any Dyson-type capture; deepening null tightens the ceiling on Phase 4–5 throughput (currently $P_{\text{comp}} \lesssim 10^{3-4} L_{\odot}$)
T3 †	GW signature inconsistent with single point mass	stochastic background or multiple low-mass inspirals from core	LISA	late 2030s–2040s	H_2 entirely (confirms H_1); removes P2 substrate
T4	EMRI point mass too small	clean chirp, $M < 5 \times 10^3 M_{\odot}$ at $> 3\sigma$	LISA	late 2030s–2040s	The selection ranking of Section 3.1: ω Cen would no longer be a P2-optimal destination, though a $5 \times 10^3 M_{\odot}$ hole still delivers $\sim 6 \times 10^{34}$ W (Eq. 2), so H_2 is weakened rather than excluded
T5 †	EMRI spin measurement low	$a_{\star} < 0.1$ at $> 3\sigma$	LISA	late 2030s–2040s	The exploitation model: no BZ reservoir, no spin-up history; MTH survives only in unmodified “pre-arrival” form
T6	High-energy neutrino bursts from core	≥ 3 tracks within 1° in 10^{2-3} s, $E \gtrsim 10$ TeV, $> 5\sigma$ post-trials, no astrophysical counterpart	KM3NeT/ARCA (ω Cen is up-going from the Mediterranean; dec. -47°)	2026–2035	A null at full sensitivity closes the Dvali and Osmanov (2023) channel; a detection is consistent with (not proof of) H_2

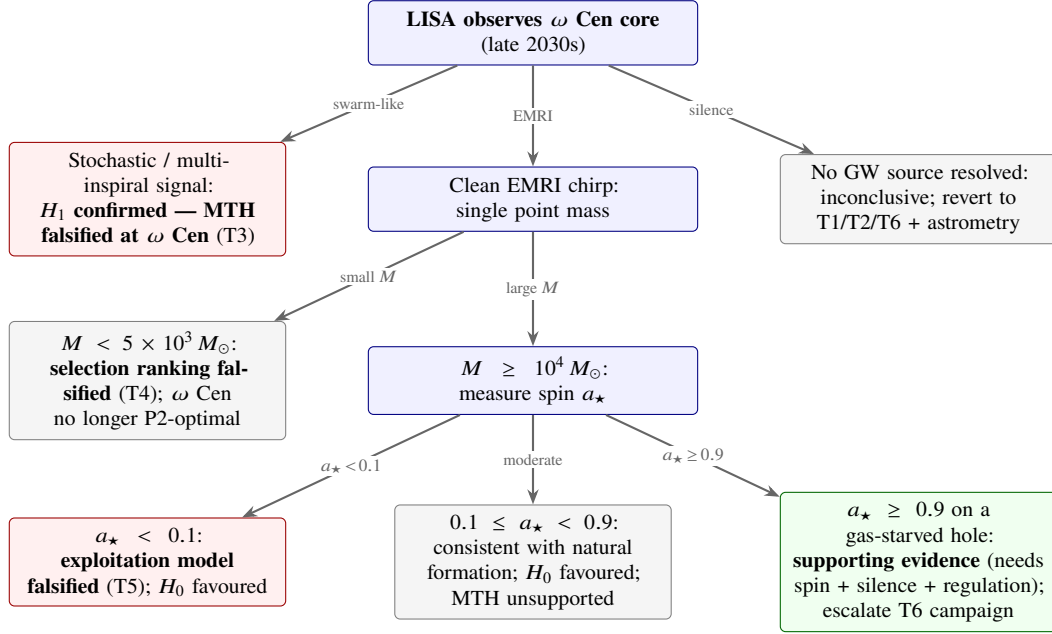


Figure 6: Decision tree for the decisive (gravitational-wave) branch of the falsification program. Red boxes terminate the hypothesis or its exploitation model; the single green branch is the only outcome in which the MTH gains material support, and even there the inference is to “anomaly requiring explanation,” not detection.

areas and atmospheric fluxes from the KM3NeT 2.0 letter of intent; [Adrián-Martínez et al., 2016](#)). A 1° -radius cone subtends $\sim 10^{-3}$ sr, a fraction $\sim 1.5 \times 10^{-4}$ of the up-going hemisphere, giving a source-cone rate of ~ 0.15 events yr^{-1} , i.e. $\lambda \sim 5 \times 10^{-9} \text{ s}^{-1}$. The Poisson expectation in a 10^3 s window is then $\mu \sim 5 \times 10^{-6}$, so the probability of an accidental ≥ 3 -fold multiplet per window is $\mu^3/6 \sim 2 \times 10^{-17}$; a decade of livetime contains $\sim 3 \times 10^5$ independent 10^3 s windows, for an expected false multiplet count of $\sim 10^{-11}$ per decade. The 1° cone is itself deliberately conservative: ARCA’s track angular resolution above 10 TeV is $\sim 0.1\text{--}0.2^\circ$, so a resolution-matched cone carries a background lower by a further factor of 25–100. The criterion is background-free by many orders of magnitude even allowing generous uncertainties in the effective area and a trials factor for scanning window durations; the practical limit on T6 is exposure, not background. A dedicated ω Cen technosignature campaign would be the first of its kind for any globular cluster beyond the pilot survey of [Huang et al. \(2026b\)](#), which could not observe ω Cen.

7.1 Supporting evidence criteria

Symmetry requires stating the confirmation side. The MTH gains support only from conjunctions that H_0 finds awkward: (i) a LISA-confirmed IMBH with $a_* \gtrsim 0.9$ in conjunction with Gyr-scale gas starvation and the regulation signature of T1, since high spin alone admits the natal-disk explanation above; (ii) repeated $> 5\sigma$ neutrino bursts from the core with hard spectra and no electromagnetic counterpart, matching the [Dvali and Osmanov \(2023\)](#) phenomenology; or (iii) a measured low-mass mass-function slope in the central 0.1 pc lying below the [González Prieto et al. \(2025\)](#) model envelope at $> 3\sigma$ (oMEGACat photometry reaches the relevant depth), the concrete form of the P4c depletion residue. No single line, including all three jointly, would constitute proof of technology; they would constitute an anomaly stack justifying escalated scrutiny. Conversely, we commit in advance: if LISA delivers T3, T4, or T5, the ω Cen application of the MTH should be retired without special pleading.

8 Discussion: objections and limits

Parsimony. The MTH posits extraterrestrial intelligence; H_0 posits nothing. By any reasonable prior the null is favoured, and we have structured every comparison accordingly. The defensible claim is that the MTH is cheap to test relative to its information value, not that it is probable: it concentrates the diffuse question “where is everybody?” onto a handful of named objects and a decade-scale instrument schedule, and its decisive observables (IMBH reality, mass, spin) are independently first-rank astrophysics that will be measured anyway.

The anthropic objection. If P1–P2 captured all civilizations, our own existence as a young, loud, planet-bound species is unremarkable; the MTH explains the silence of the old, not the existence of the young. It is therefore not undermined by our own counterexample, but neither can it explain why no expansionist civilizations are visible (cf. [Hanson et al. 2021](#)): if migration is optional, loud lineages should still exist somewhere. The defensible position is that the MTH thins the expected population of loud civilizations by removing its most capable members, sharpening rather than fully dissolving the paradox; full dissolution requires either high migration compliance or supplementary rarity from conventional Drake-equation factors.

Goal stability over 10^8 years. Phase 4 presupposes coherent purpose across timescales that dwarf all institutional experience. We flagged this as the architecture’s largest unargued premise (Section 6). One partial mitigation: the architecture’s payoff structure is front-loaded (Phase 3 already yields a substrate orders of magnitude beyond planetary computing), so a lineage that fragments mid-program still leaves durable traces, but fewer than we once supposed. The companion analysis ([Swanson, 2026](#)) finds that an abandoned swarm is not preserved: collisional cascade grinds it to debris within 10^{2-3} yr, and the debris drains into the hole. The durable residues of incoherence are the spin acquired before fragmentation and, if Phase 4 ran long enough, the core depletion it caused; the hardware itself leaves no fossil. The observational program is robust to civilizational incoherence only through those two channels, which is a further reason the dynamical tests (T3–T5) carry the weight.

Tidal and radiative survivability at the ISCO. At $2 \times 10^4 M_\odot$ the tidal field at the Schwarzschild ISCO is $2GM/r_{\text{ISCO}}^3 \simeq 1 \text{ s}^{-2}$: a 10-m node experiences a benign $\sim 1 g$ differential, while kilometre-scale rigid structures would face $\sim 10^2 g$ — one reason the architecture assumes a swarm of small free-flying nodes rather than a monolithic platform. Radiation from even a managed accretion disk is the harsher constraint, and motivates the tiered-orbit design and sub-Eddington feeding discipline assumed in Section 6. We have not engineered these systems in detail and do not pretend to; the claim defended is the thermodynamic gradient (P2), not any specific machine.

Why has nothing arrived here? A migration hypothesis must explain why migrators’ probes are not conspicuous in every system, reviving [Hart \(1975\)](#) and [Tipler \(1980\)](#). The MTH answer is economic: for a computation-maximizer, planetary systems are at most transit infrastructure rather than destinations; the Galaxy’s uncollected starlight (an integrated stellar luminosity of $(1-2) \times 10^{37}$ W) dwarfs any single system’s output by ten orders of magnitude, and yet (Section 4.5) the marginal value of any outward claim is dominated by deepening the home gravity well. Quiet, minimal, purpose-built transit, rather than occupation, is the predicted footprint, consistent with the null results of artefact SETI to date ([Wright et al., 2022](#)). [Ivliev \(2026\)](#) presses the opposite conclusion: once autonomous machine industry matures, outward expansion becomes too cheap and too useful for any civilization to refuse, and the attractor is

quiet distributed expansion rather than concentration. We accept the premise that expansion is cheap, and we concede that for an undiscounted maximizer of total integrated computation Ivliev’s inference goes through, since controlled mass grows as t^3 . The disagreement localizes to the auxiliary assumptions of Section 4.5: under temporal discounting, latency-bound unified computation, or bounded planning horizons, expansion does not compete with concentration in yield per unit of controlled resource, and it is civilizations satisfying at least one of those conditions that the MTH captures. The two proposals are nonetheless observationally adjacent, since both predict a Galaxy free of conspicuous macro-engineering, and they diverge where the MTH is testable: Ivliev’s filter predicts no privileged loci, while the MTH concentrates its residues on a short list of named systems.

The kugelblitz weak link. The only positive high-energy signature we predict (T6) inherits the controversy over micro-black-hole formation from radiation (Álvarez-Domínguez et al., 2024; Loeb, 2024). We therefore weight T6 as the cheapest test, not the strongest: its null closes a speculative channel; its detection would be extraordinary but stands or falls with Dvali and Osmanov (2023) physics, not with the MTH core.

Scope of the ω Cen bet. The MTH is a selection theory over a class; ω Cen is its highest-ranked accessible member, not its definition. Falsification at ω Cen (T3, or the ranking demotion of T4) retires the flagship application and substantially weakens the hypothesis (the Galaxy’s best candidate failing is evidence about the class), but formally the theory survives in 47 Tuc, M54, and extragalactic nuclear clusters. We accept the methodological hazard this creates (unfalsifiability by retreat) and therefore bind ourselves to the class-level prediction: *if no Galactic globular-cluster IMBH with $a_\star \gtrsim 0.9$ exists, the MTH is wrong for the Milky Way*, a statement LISA-era gravitational-wave astronomy can settle.

9 Conclusion

We have argued, first as a matter of established physics, that rapidly spinning massive black holes in dense old stellar clusters constitute the global optimum among present-day environments for long-term computation (by factors of $\sim 10^2$ in energy per unit fuel, $\sim 10^9$ in continuous power, a realized 10^6 – 10^9 in entropy-disposal cost once delivery overheads are charged against the $\sim 10^{12}$ floor-to-floor ratio (Section 4.2), and unbounded factors in archival density), and second, as an explicitly speculative hypothesis, that this gradient acts on the oldest technological civilizations as an attractor whose operation would look, from outside, like the silence we observe. The Macro Transcension Hypothesis inherits the thermodynamic insight of transcension and aestivation while repairing their respective defects: it needs no exit from the universe and no dormancy, and it survives the Bennett et al. (2019) free-energy objection by importing the cold reservoir into the present rather than waiting for one.

Its principal virtue is observational specificity. The hypothesis selects a named target (Omega Centauri, whose contested 8×10^3 – $5 \times 10^4 M_\odot$ central object, perfect electromagnetic silence, and 12-Gyr head start make it scientifically urgent on entirely conventional grounds) and commits to kill criteria that existing and funded instruments will adjudicate: LISA’s resolution of the IMBH-versus-swarm question and, above all, its spin measurement; deep accretion and waste-heat limits from JWST and southern radio arrays; and a first-of-its-kind neutrino monitoring campaign with KM3NeT. If the central object proves to be a remnant swarm, a light point mass, or a slowly spinning hole, the hypothesis fails at its flagship and we have said so in advance. If instead the 2030s deliver a massive, gas-starved, near-extremally spinning black hole in the quietest old cluster in the sky, the question this paper poses will have earned the right to be taken seriously.

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Data availability

No new observational data were generated. All cited measurements are available in the referenced publications.

References

- Aartsen, M. G. et al. (2020). Time-integrated neutrino source searches with 10 years of IceCube data. *Physical Review Letters*, 124:051103.
- Adrián-Martínez, S. et al. (2016). Letter of intent for KM3NeT 2.0. *Journal of Physics G: Nuclear and Particle Physics*, 43(8):084001.
- Álvarez-Domínguez, Á., Garay, L. J., Martín-Martínez, E., and Polo-Gómez, J. (2024). No black holes from light. *Physical Review Letters*, 133:041401.
- Amaro-Seoane, P. (2018). Relativistic dynamics and extreme mass ratio inspirals. *Living Reviews in Relativity*, 21:4.
- Armstrong, S. and Sandberg, A. (2013). Eternity in six hours: Intergalactic spreading of intelligent life and sharpening the Fermi paradox. *Acta Astronautica*, 89:1–13.
- Athas, W. C., Svensson, L. J., Koller, J. G., Tzartzanis, N., and Chou, E. Y.-C. (1994). Low-power digital systems based on adiabatic-switching principles. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2(4):398–407.
- Babak, S., Gair, J., Sesana, A., Barausse, E., Sopuerta, C. F., Berry, C. P. L., Berti, E., Amaro-Seoane, P., Petiteau, A., and Klein, A. (2017). Science with the space-based interferometer LISA. V. Extreme mass-ratio inspirals. *Physical Review D*, 95(10):103012.
- Bañares-Hernández, A., Calore, F., Martín Camalich, J., and Read, J. I. (2025). New constraints on the central mass contents of Omega Centauri from combined stellar kinematics and pulsar timing. *Astronomy & Astrophysics*, 693:A104.
- Bardeen, J. M. (1970). Kerr metric black holes. *Nature*, 226:64–65.
- Bardeen, J. M., Press, W. H., and Teukolsky, S. A. (1972). Rotating black holes: locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *The Astrophysical Journal*, 178:347–370.

- Barrow, J. D. (1998). *Impossibility: The Limits of Science and the Science of Limits*. Oxford University Press, Oxford.
- Baumgardt, H., He, C., Sweet, S. M., Drinkwater, M., Sollima, A., Hurley, J., Usher, C., Kamann, S., Dalglish, H., Dreizler, S., and Husser, T.-O. (2019). No evidence for intermediate-mass black holes in the globular clusters ω Cen and NGC 6624. *Monthly Notices of the Royal Astronomical Society*, 488:5340–5351.
- Baumgardt, H. and Vasiliev, E. (2021). Accurate distances to Galactic globular clusters through a combination of Gaia EDR3, HST, and literature data. *Monthly Notices of the Royal Astronomical Society*, 505:5957–5977.
- Bekenstein, J. D. (1973). Black holes and entropy. *Physical Review D*, 7:2333–2346.
- Bekenstein, J. D. (1974). Generalized second law of thermodynamics in black-hole physics. *Physical Review D*, 9:3292–3300.
- Bekenstein, J. D. (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. *Physical Review D*, 23:287–298.
- Bekki, K. and Tsujimoto, T. (2019). A new formation model for ω Centauri: A complex interplay of astrophysical processes. *The Astrophysical Journal*, 886:121.
- Bennett, C. H. (1973). Logical reversibility of computation. *IBM Journal of Research and Development*, 17:525–532.
- Bennett, C. H. (1982). The thermodynamics of computation—a review. *International Journal of Theoretical Physics*, 21:905–940.
- Bennett, C. H., Hanson, R., and Riedel, C. J. (2019). Comment on ‘The aestivation hypothesis for resolving Fermi’s paradox’. *Foundations of Physics*, 49:820–829.
- Benson-Tilsen, T. and Soares, N. (2016). Formalizing convergent instrumental goals. In *AAAI Workshop on AI, Ethics, and Society*.
- Bérut, A., Arakelyan, A., Petrosyan, A., Ciliberto, S., Dillenschneider, R., and Lutz, E. (2012). Experimental verification of Landauer’s principle linking information and thermodynamics. *Nature*, 483:187–189.
- Blandford, R. D. and Znajek, R. L. (1977). Electromagnetic extraction of energy from Kerr black holes. *Monthly Notices of the Royal Astronomical Society*, 179:433–456.
- Bostrom, N. (2012). The superintelligent will: Motivation and instrumental rationality in advanced artificial agents. *Minds and Machines*, 22:71–85.
- Bousso, R. (2002). The holographic principle. *Reviews of Modern Physics*, 74:825–874.
- Bradbury, R. J. (1999). Matrioshka brains. Unpublished working manuscript, revised through 2001. Archived at <https://gwern.net/doc/ai/scaling/hardware/1999-bradbury-matrioshkabrains.pdf>; mirrored at <https://web.archive.org/web/2024/https://gwern.net/doc/ai/scaling/hardware/1999-bradbury-matrioshkabrains.pdf>.
- Breen, P. G. and Heggie, D. C. (2013). Dynamical evolution of black hole subsystems in idealized star clusters. *Monthly Notices of the Royal Astronomical Society*, 432:2779–2797.

- Breit, G. and Wheeler, J. A. (1934). Collision of two light quanta. *Physical Review*, 46:1087–1091.
- Camilloni, F. and Rezzolla, L. (2025). Self-consistent multidimensional Penrose process driven by magnetic reconnection. *The Astrophysical Journal Letters*, 982:L31.
- Casini, H. (2008). Relative entropy and the Bekenstein bound. *Classical and Quantum Gravity*, 25:205021.
- Chen, S., Hare, J., Kargaltsev, O., Yang, H., Cioffi, D., Häberle, M., and Seth, A. (2025). The intermediate mass black hole in Omega Centauri: Constraints on accretion from JWST. Submitted to ApJ.
- Christodoulou, D. (1970). Reversible and irreversible transformations in black-hole physics. *Physical Review Letters*, 25:1596–1597.
- Christodoulou, D. and Ruffini, R. (1971). Reversible transformations of a charged black hole. *Physical Review D*, 4:3552–3555.
- Ćirković, M. M. (2018). *The Great Silence: The Science and Philosophy of Fermi’s Paradox*. Oxford University Press, Oxford.
- Ćirković, M. M. and Bradbury, R. J. (2006). Galactic gradients, postbiological evolution and the apparent failure of SETI. *New Astronomy*, 11(8):628–639.
- Clontz, C., Seth, A. C., Dotter, A., Häberle, M., Neumayer, N., et al. (2024). oMEGACat. IV. Constraining the ages of Omega Centauri subgiant branch stars with HST and MUSE. *The Astrophysical Journal*, 977:14.
- Colom i Bernadich, M., Dai, S., Abbate, F., Kerr, M., Bachetti, M., Buchner, S., Johnston, S., Burgay, M., Possenti, A., Ridolfi, A., Freire, P. C. C., Kramer, M., Stappers, B., et al. (2026). A joint MeerKAT and Parkes view of Omega Centauri: New TRAPUM searches and pulsar timing. *arXiv e-prints*, page arXiv:2603.21845. Under review.
- Colpi, M. et al. (2024). LISA definition study report. ESA-SCI-DIR-RP-002.
- Comisso, L. and Asenjo, F. A. (2021). Magnetic reconnection as a mechanism for energy extraction from rotating black holes. *Physical Review D*, 103:023014.
- Criswell, D. R. (1985). Solar system industrialization: Implications for interstellar migrations. In Finney, B. R. and Jones, E. M., editors, *Interstellar Migration and the Human Experience*, pages 50–87. University of California Press, Berkeley.
- Curtis, O., Adams, V. H., Angerhausen, D., Bates, J., Berea, A., Dick, S. J., Elvis, M., Khatri, S. P., Linares, R., Muco, M., Seager, S., and Wright, J. T. (2026). The Dyson Minds 2025 workshop: SETI around black holes. *Publications of the Astronomical Society of the Pacific*, 138:046001.
- Dvali, G. and Osmanov, Z. N. (2023). Black holes as tools for quantum computing by advanced extraterrestrial civilizations. *International Journal of Astrobiology*, 22(6):617–640.
- Dyson, F. J. (1960). Search for artificial stellar sources of infrared radiation. *Science*, 131(3414):1667–1668.
- Event Horizon Telescope Collaboration (2022). First Sagittarius A* Event Horizon Telescope results. I. The shadow of the supermassive black hole in the center of the Milky Way. *The Astrophysical Journal Letters*, 930:L12.

- Forgan, D. H. (2019). *Solving Fermi’s Paradox*. Cambridge Astrobiology. Cambridge University Press, Cambridge.
- Frank, J., King, A., and Raine, D. (2002). *Accretion Power in Astrophysics*. Cambridge University Press, Cambridge, 3rd edition.
- Frank, M. P. (2002). The physical limits of computing. *Computing in Science & Engineering*, 4(3):16–26.
- Fredkin, E. and Toffoli, T. (1982). Conservative logic. *International Journal of Theoretical Physics*, 21:219–253.
- Freitas, Jr., R. A. and Gilbreath, W. P., editors (1982). *Advanced Automation for Space Missions*. Number 2255 in NASA Conference Publication. NASA. Edited volume; proceedings of the 1980 NASA/ASEE Summer Study, University of Santa Clara.
- Frolov, V. P. and Novikov, I. D. (1998). *Black Hole Physics: Basic Concepts and New Developments*. Kluwer Academic, Dordrecht.
- Fujii, M. S., Wang, L., Tanikawa, A., Hirai, Y., and Saitoh, T. R. (2024). Simulations predict intermediate-mass black hole formation in globular clusters. *Science*, 384:1488–1492.
- González Prieto, E., Rodríguez, C. L., and Cabrera, T. (2025). Growing the intermediate-mass black hole in Omega Centauri. *The Astrophysical Journal Letters*, 990:L69.
- GRAVITY Collaboration (2022). Mass distribution in the Galactic Center based on interferometric astrometry of multiple stellar orbits. *Astronomy & Astrophysics*, 657:L12.
- Greene, J. E., Strader, J., and Ho, L. C. (2020). Intermediate-mass black holes. *Annual Review of Astronomy and Astrophysics*, 58:257–312.
- Griffith, R. L., Wright, J. T., Maldonado, J., Povich, M. S., Sigurdsson, S., and Mullan, B. (2015). The $\hat{g}$ infrared search for extraterrestrial civilizations with large energy supplies. III. The reddest extended sources in WISE. *The Astrophysical Journal Supplement Series*, 217(2):25.
- Häberle, M., Neumayer, N., Bellini, A., Libralato, M., Clontz, C., Seth, A. C., et al. (2024a). oMEGACat. II. Photometry and proper motions for 1.4 million stars in Omega Centauri and its rotation in the plane of the sky. *The Astrophysical Journal*, 970:192.
- Häberle, M., Neumayer, N., Clontz, C., Seth, A. C., Bellini, A., et al. (2025). oMEGACat. VI. Analysis of the overall kinematics of Omega Centauri in 3D: Velocity dispersion, kinematic distance, anisotropy, and energy equipartition. *The Astrophysical Journal*, 983:95.
- Häberle, M., Neumayer, N., Seth, A., Bellini, A., Libralato, M., Anderson, J., Whitaker, M., van de Ven, G., Watkins, L. L., Clontz, C., et al. (2024b). Fast-moving stars around an intermediate-mass black hole in ω Centauri. *Nature*, 631(8020):285–289.
- Haggard, D., Cool, A. M., Heinke, C. O., van der Marel, R., Cohn, H. N., Lugger, P. M., and Anderson, J. (2013). A deep Chandra X-ray limit on the putative IMBH in Omega Centauri. *The Astrophysical Journal Letters*, 773:L31.
- Hanson, R., Martin, D., McCarter, C., and Paulson, J. (2021). If loud aliens explain human earliness, quiet aliens are also rare. *The Astrophysical Journal*, 922(2):182.

- Harris, W. E. (1996). A catalog of parameters for globular clusters in the Milky Way. *The Astronomical Journal*, 112:1487. 2010 edition: arXiv:1012.3224.
- Hart, M. H. (1975). An explanation for the absence of extraterrestrials on Earth. *Quarterly Journal of the Royal Astronomical Society*, 16:128–135.
- Hawking, S. W. (1974). Black hole explosions? *Nature*, 248:30–31.
- Hawking, S. W. (1975). Particle creation by black holes. *Communications in Mathematical Physics*, 43:199–220.
- Hilker, M. and Richtler, T. (2000). ω Centauri – a former nucleus of a dissolved dwarf galaxy? New evidence from Strömgen photometry. *Astronomy & Astrophysics*, 362:895–909.
- Hsiao, T. Y.-Y., Goto, T., Hashimoto, T., Santos, D. J. D., On, A. Y. L., Kilerci-Eser, E., Wong, Y. H. V., Kim, S. J., Wu, C. K.-W., Ho, S. C.-C., and Lu, T.-Y. (2021). A dyson sphere around a black hole. *Monthly Notices of the Royal Astronomical Society*, 506(2):1723–1732.
- Huang, B.-L., Tao, Z.-Z., and Zhang, T.-J. (2026a). Phase-space crystallization in Galactic globular clusters: A Gaia-based metric and implications for technosignature searches. *The Astrophysical Journal*. Accepted.
- Huang, B.-L., Tao, Z.-Z., Zhang, T.-J., and Gajjar, V. (2026b). The FAST-SETI Milky Way globular cluster survey. I. A pilot multibeam on-the-fly search of five globular clusters at the L band. *The Astronomical Journal*, 171(1):51.
- Ibata, R. A., Bellazzini, M., Malhan, K., Martin, N., and Bianchini, P. (2019). Identification of the long stellar stream of the prototypical massive globular cluster ω Centauri. *Nature Astronomy*, 3:667–672.
- Inoue, M. and Yokoo, H. (2011). Type III dyson sphere of highly advanced civilisations around a super massive black hole. *Journal of the British Interplanetary Society*, 64:58–62.
- Ivliev, S. (2026). Autonomous AI-cosmoindustry and the quiet expansion filter: A threshold-based resolution of the Fermi paradox.
- Johnson, C. I. and Pilachowski, C. A. (2010). Chemical abundances for 855 giants in the globular cluster Omega Centauri (NGC 5139). *The Astrophysical Journal*, 722:1373–1410.
- Kardashev, N. S. (1964). Transmission of information by extraterrestrial civilizations. *Soviet Astronomy*, 8:217–221.
- Kızıltan, B., Baumgardt, H., and Loeb, A. (2017). An intermediate-mass black hole in the centre of the globular cluster 47 Tucanae. *Nature*, 542:203–205.
- Lacki, B. C. and DiKerby, S. (2025). Possibilities for SETI at high energy. White paper submitted to the 2025 NASA DARES RFI.
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5:183–191.
- Lingam, M. and Loeb, A. (2021). *Life in the Cosmos: From Biosignatures to Technosignatures*. Harvard University Press, Cambridge, MA.

- Lloyd, S. (2000). Ultimate physical limits to computation. *Nature*, 406:1047–1054.
- Loeb, A. (2024). Comment on “No black holes from light”. Preprint; reply by original authors at arXiv:2408.11097.
- Lubin, P. (2016). A roadmap to interstellar flight. *Journal of the British Interplanetary Society*, 69:40–72.
- Mahida, A. D. et al. (2026). No evidence for accretion around the intermediate-mass black hole in Omega Centauri. *The Astrophysical Journal*, 996:122.
- Mann, C. R., Richer, H., Heyl, J., Anderson, J., Kalirai, J., Caiazzo, I., Möhle, S., Knee, A., and Baumgardt, H. (2019). A multi-mass velocity dispersion model of 47 Tucanae indicates no evidence for an intermediate mass black hole. *The Astrophysical Journal*, 875:1.
- Margolus, N. and Levitin, L. B. (1998). The maximum speed of dynamical evolution. *Physica D: Nonlinear Phenomena*, 120:188–195.
- Martinez, M. A. S., González Prieto, E., and Rasio, F. A. (2026). Survival analysis of intermediate-mass black holes in dense star clusters. *arXiv e-prints*, page arXiv:2602.23431. Submitted to ApJ.
- Matloff, G. L. (2017). Star lifting: An application for alien megastructures. *Journal of the British Interplanetary Society*, 70:458–460.
- Meringolo, C., Camilloni, F., and Rezzolla, L. (2025). Electromagnetic energy extraction from Kerr black holes: Ab initio calculations. *The Astrophysical Journal Letters*, 992:L8.
- Misner, C. W., Thorne, K. S., and Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman, San Francisco.
- Nathanail, A. (2026). Black hole limits redefined: Extreme efficiency in black hole jets. Preprint, not yet peer-reviewed.
- Novikov, I. D. and Thorne, K. S. (1973). Astrophysics of black holes. In DeWitt, C. and DeWitt, B. S., editors, *Black Holes (Les Houches)*, pages 343–450. Gordon and Breach, New York.
- Noyola, E., Gebhardt, K., and Bergmann, M. (2008). Gemini and Hubble Space Telescope evidence for an intermediate-mass black hole in ω Centauri. *The Astrophysical Journal*, 676:1008–1015.
- Omohundro, S. M. (2008). The basic ai drives. In Wang, P., Goertzel, B., and Franklin, S., editors, *Proceedings of the First AGI Conference (AGI-08)*, pages 483–492. IOS Press.
- Opatrný, T., Richterek, L., and Bakala, P. (2017). Life under a black sun. *American Journal of Physics*, 85(1):14–22.
- Page, D. N. (1976). Particle emission rates from a black hole: Massless particles from an uncharged, nonrotating hole. *Physical Review D*, 13:198–206.
- Parrondo, J. M. R., Horowitz, J. M., and Sagawa, T. (2015). Thermodynamics of information. *Nature Physics*, 11:131–139.
- Penrose, R. (1969). Gravitational collapse: the role of general relativity. *Rivista del Nuovo Cimento*, 1(Numero Speciale):252–276. Reprinted: Gen. Relativ. Gravit. 34, 1141 (2002).

- Penrose, R. and Floyd, R. M. (1971). Extraction of rotational energy from a black hole. *Nature Physical Science*, 229:177–179.
- Pryor, C. and Meylan, G. (1993). Velocity dispersions for Galactic globular clusters. In Djorgovski, S. G. and Meylan, G., editors, *Structure and Dynamics of Globular Clusters*, volume 50 of *ASP Conference Series*, page 357. Astronomical Society of the Pacific.
- Rauch, K. P. and Tremaine, S. (1996). Resonant relaxation in stellar systems. *New Astronomy*, 1:149–170.
- Reynolds, C. S. (2021). Observational constraints on black hole spin. *Annual Review of Astronomy and Astrophysics*, 59:117–154.
- Sandberg, A., Armstrong, S., and Ćirković, M. M. (2016). That is not dead which can eternal lie: the aestivation hypothesis for resolving Fermi’s paradox. *Journal of the British Interplanetary Society*, 69(11):406–415.
- Schwinger, J. (1951). On gauge invariance and vacuum polarization. *Physical Review*, 82:664–679.
- Scoggins, M. T. and Kipping, D. (2023). Lazarus stars: numerical investigations of stellar evolution with star-lifting as a life extension strategy. *Monthly Notices of the Royal Astronomical Society*, 523:3251–3257.
- Shapiro, S. L. and Teukolsky, S. A. (1983). *Black Holes, White Dwarfs, and Neutron Stars: The Physics of Compact Objects*. Wiley-Interscience, New York.
- Smart, J. M. (2012). The transcension hypothesis: Sufficiently advanced civilizations invariably leave our universe, and implications for METI and SETI. *Acta Astronautica*, 78:55–68.
- Socas-Navarro, H., Haqq-Misra, J., Wright, J. T., Kopparapu, R., Benford, J., Davis, R., and TechnoClimes 2020 workshop participants (2021). Concepts for future missions to search for technosignatures. *Acta Astronautica*, 182:446–453.
- Soltis, J., Casertano, S., and Riess, A. G. (2021). The parallax of ω Centauri measured from Gaia EDR3 and a direct, geometric calibration of the tip of the red giant branch and the Hubble constant. *The Astrophysical Journal Letters*, 908:L5.
- Souza, S., Neumayer, N., Seth, A. C., Wang, Z., Clontz, C., Häberle, M., Nitschai, M. S., Smith, P. J., Matsuno, T., et al. (2026). oMEGACat. X. shedding light on the disrupted dwarf galaxy of Omega Centauri. *arXiv e-prints*, page arXiv:2603.23589. Submitted to ApJ.
- Strokov, V., Fragione, G., Wong, K. W. K., Helfer, T., and Berti, E. (2022). Hunting for intermediate-mass black holes with LISA binary radial velocity measurements. *Physical Review D*, 105:124048.
- Swanson, T. (2026). Engineered intermediate-mass black hole systems: Infrastructure constraints, observable residue, and a multi-messenger adjudication framework. Preprint; omegacentauri.me.
- Tarter, J. (2001). The search for extraterrestrial intelligence (SETI). *Annual Review of Astronomy and Astrophysics*, 39:511–548.
- Tchekhovskoy, A. (2015). Launching of active galactic nuclei jets. In Contopoulos, I., Gabuzda, D., and Kylafis, N., editors, *The Formation and Disruption of Black Hole Jets*, volume 414 of *Astrophysics and Space Science Library*, pages 45–82. Springer, Cham.

- Tchekhovskoy, A., Narayan, R., and McKinney, J. C. (2011). Efficient generation of jets from magnetically arrested accretion on a rapidly spinning black hole. *Monthly Notices of the Royal Astronomical Society: Letters*, 418:L79–L83.
- The KM3NeT Collaboration (2025). Observation of an ultra-high-energy cosmic neutrino with KM3NeT. *Nature*, 638(8050):376–382.
- Thorne, K. S. (1974). Disk-accretion onto a black hole. II. Evolution of the hole. *The Astrophysical Journal*, 191:507–519.
- Thorne, K. S. (1994). *Black Holes and Time Warps: Einstein's Outrageous Legacy*. W. W. Norton, New York.
- Thorne, K. S., Price, R. H., and Macdonald, D. A. (1986). *Black Holes: The Membrane Paradigm*. Yale University Press, New Haven.
- Tipler, F. J. (1980). Extraterrestrial intelligent beings do not exist. *Quarterly Journal of the Royal Astronomical Society*, 21:267–281.
- Toffoli, T. (1980). Reversible computing. In de Bakker, J. and van Leeuwen, J., editors, *Automata, Languages and Programming (ICALP 1980)*, volume 85 of *Lecture Notes in Computer Science*, pages 632–644, Berlin, Heidelberg. Springer.
- Tremou, E., Strader, J., Chomiuk, L., Shishkovsky, L., Maccarone, T. J., Miller-Jones, J. C. A., et al. (2018). The MAVERIC survey: Still no evidence for accreting intermediate-mass black holes in globular clusters. *The Astrophysical Journal*, 862:16.
- van der Marel, R. P. and Anderson, J. (2010). New limits on an intermediate-mass black hole in Omega Centauri. II. Dynamical models. *The Astrophysical Journal*, 710:1063–1088.
- Vidal, C. (2011). Black holes: Attractors for intelligence? Presented at the Kavli Royal Society International Centre, “Towards a scientific and societal agenda on extra-terrestrial life”.
- Vidal, C. (2014). *The Beginning and the End: The Meaning of Life in a Cosmological Perspective*. The Frontiers Collection. Springer, Cham.
- Webb, S. (2015). *If the Universe Is Teeming with Aliens... Where Is Everybody? Seventy-Five Solutions to the Fermi Paradox and the Problem of Extraterrestrial Life*. Science and Fiction. Springer, Cham, 2 edition.
- Whitaker, M., Kerr, E., Seth, A., Häberle, M., Strader, J., Anderson, J., Bellini, A., Clontz, C., Freeman, Z., Griggio, M., Kamann, S., Libralato, M., Neumayer, N., González Prieto, E., Rodríguez, C. L., Saracino, S., Smith, P., van de Ven, G., and Wang, Z. (2026). A long period stellar-mass black hole binary in ω Centauri. *The Astrophysical Journal Letters*.
- Wolpert, D. H. (2019). The stochastic thermodynamics of computation. *Journal of Physics A: Mathematical and Theoretical*, 52:193001.
- Wright, J. T., Haqq-Misra, J., Frank, A., Kopparapu, R., Lingam, M., and Sheikh, S. Z. (2022). The case for technosignatures: Why they may be abundant, long-lived, highly detectable, and unambiguous. *The Astrophysical Journal Letters*, 927(2):L30.

- Wright, J. T., Mullan, B., Sigurdsson, S., and Povich, M. S. (2014). The $\hat{g}$ infrared search for extraterrestrial civilizations with large energy supplies. I. Background and justification. *The Astrophysical Journal*, 792(1):26.
- Zocchi, A., Gieles, M., and Hénault-Brunet, V. (2019). The effect of stellar-mass black holes on the central kinematics of ω Cen: a cautionary tale for IMBH interpretations. *Monthly Notices of the Royal Astronomical Society*, 482:4713–4725.